Effects of Subsidies on Welfare and Market Structure in the U.S. Broadband Industry*

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Abstract

How should governments split broadband subsidies between consumer discounts and infrastructure grants? I study the Affordable Connectivity Program (ACP) and the Broadband Equity, Access, and Deployment (BEAD) program using state-year event studies and a tract-level structural model with endogenous entry. ACP was followed by lower prices, faster plan offerings, and lower concentration in high-eligibility states. BEAD produced smaller, delayed effects, consistent with construction and entry lags. I then use FCC Form 477, Urban Rate Survey, and American Community Survey data to estimate demand and entry for 220,814 products in 69,085 tracts. Counterfactuals imply a BEAD benefit-cost ratio of 3.5–12.7 and an ACP ratio of 1.66–4.04. ACP is most valuable where broadband exists but remains unaffordable; BEAD is most valuable where infrastructure is missing. The programs are complements, not substitutes.

JEL classification: D22, L13, L51, L52, L96, L98.

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1 Introduction

Broadband access has become a prerequisite for economic participation, yet the digital divide remains wide. In 2020, roughly 21 million American households lacked fixed broadband service, a gap that falls disproportionately on low-income and rural communities.[1] Congress responded with the Infrastructure Investment and Jobs Act (IIJA) of 2021, which allocated approximately $65 billion to broadband and created two flagship programs: the Affordable Connectivity Program (ACP), a $14.2 billion demand-side subsidy providing eligible low-income households with monthly discounts of up to $30 on broadband bills, and the Broadband Equity, Access, and Deployment (BEAD) program, a $42.5 billion supply-side grant funding deployment in unserved and underserved areas. The two programs share a goal—wider, more affordable broadband—but attack different constraints. ACP lowers the consumer-facing price where service already exists. BEAD lowers the fixed cost of building service where private deployment has not materialized.

In this paper, I ask three questions. First, did ACP and BEAD change broadband prices, product offerings, and local market structure after IIJA? Second, how large are the welfare gains from each program once firms can enter or adjust local product positions? Third, are demand-side and supply-side subsidies substitutes, or do they address different parts of the digital divide? These questions matter for current policy. ACP exhausted its funding in June 2024, leaving policymakers to decide whether a successor affordability program is worth financing. BEAD is the largest federal broadband deployment program in U.S. history, but its returns depend on whether subsidies induce entry in markets where deployment is privately unprofitable. Standard evaluations that count new subscribers or dollars per connection do not answer these questions, especially in local broadband markets with few providers and substantial infra-marginal households.

To answer them, I combine reduced-form and structural evidence. The reduced-form analysis uses continuous-treatment event studies to compare states with different exposure to ACP eligibility and BEAD funding before and after IIJA. This design is well suited for measuring the timing and direction of observed policy effects. The structural model then asks what those effects imply for welfare and counterfactual policy design. I estimate a nested-logit demand system, recover Bertrand-Nash markups and marginal costs, and estimate fixed costs of entry using the scalable discrete-game bounds of Fan and Yang [2024]. This approach is appropriate here because broadband policy changes both demand and product availability. It also avoids an infeasible full solution of the entry game during estimation, which is essential in

[1] Federal Communications Commission [2020] reported 21.3 million Americans lacking access to fixed broadband at 25/3 Mbps speeds based on FCC Form 477 data (see page 19).

a setting with tens of thousands of local markets and many potential provider-speed positions.

The data combine several official federal sources. For the reduced-form analysis, I build a balanced panel of 52 U.S. states from 2014 to 2025 using FCC Urban Rate Survey prices and product characteristics, ACP eligibility exposure from the American Community Survey, and BEAD allocations from federal program records. For the structural analysis, I construct a pre-IIJA product-level dataset for December 2020. The sample contains 220,814 broadband products in 69,085 census tracts, defined at the tract-provider-speed-tier level. Product availability comes from FCC Form 477 deployment records, prices from the FCC Urban Rate Survey, adoption and outside-option shares from FCC broadband adoption data and the American Community Survey, and demographics from American Community Survey tract files. The data are unusually rich for this market, combining availability, prices, technology, demographics, adoption, and potential entry positions before either major IIJA program was implemented. To my knowledge, no existing dataset provides comparable coverage of broadband market conditions at this level of geographic and product detail.

The results show that the programs operate through different margins. In the event study, ACP is associated with lower prices, faster plan mixes, and lower concentration in high-eligibility states. Moving from the 25th to the 75th percentile of ACP eligibility exposure is associated with broadband prices that were roughly 3–5 percent lower for average plans and somewhat more for high-speed plans at peak response. BEAD effects are smaller and appear with a two-to-three year lag, consistent with the time required for infrastructure deployment. In the structural counterfactuals, a 75 percent BEAD fixed-cost subsidy induces 115,000–341,000 new provider-tract entries and raises consumer surplus from \$2.18–\$2.65 billion per month to \$2.86–\$3.36 billion. A \$30 ACP discount raises consumer surplus to \$5.31–\$5.91 billion per month by expanding take-up at lower consumer-facing prices. In cost-benefit terms, BEAD generates a BCR of 3.5–12.7, while ACP generates 1.66–4.04. The central lesson is that the programs are complements: ACP is most useful where infrastructure exists but affordability limits adoption; BEAD is most useful where infrastructure is missing.

The paper contributes to three related literatures. First, it contributes to work on broadband markets and the digital divide. Existing studies show that broadband adoption is shaped by prices, income, local market structure, and provider choice [Rosston et al., 2010, Nevo et al., 2016, Flamm and Varas, 2022, Espín and Rojas, 2024, Kearns, 2024]. I build on this evidence but study policy exposure as a source of variation in both affordability and market outcomes.[2] Second, the paper contributes to structural work on differentiated-product

[2]Concentration, high prices, and limited choice are well documented in U.S. broadband [Flamm and Varas, 2022]. Downstream consequences for labor markets, education, and health are documented in Forman et al. [2005], Goldfarb and Prince [2008], and Goolsbee and Klenow [2002].

demand and endogenous product entry. The approach is closest in spirit to Tamer [2003], Ciliberto and Tamer [2009], and Fan and Yang [2024], and brings scalable discrete-game methods to broadband markets. Third, the paper contributes to the evaluation of public subsidies in imperfectly competitive markets. I compare a recurring demand-side subsidy and a fixed-cost deployment subsidy in the same welfare accounting exercise, separating affordability from infrastructure deployment. This comparison also speaks to work on public provision and infrastructure crowd-out, including Wilson [2025], who finds that municipal broadband can reduce private investment. I do not model that crowd-out channel for BEAD, so the results should be read as the welfare return to subsidized private entry rather than to public provision.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting, data, and local market structure. Section 3 presents the reduced-form evidence. Section 4 develops the structural model, including timing assumptions, identification, and a discussion of multiple equilibria. Section 5 reports estimation results and structural parameter diagnostics. Section 6 presents counterfactual policy simulations and distributional effects across market types. Section 7 reports the cost-benefit analysis, including sensitivity to discount rate and infrastructure horizon. Section 8 concludes.

2 Industry Background and Data

2.1 Industry Background

2.1.1 Current Landscape

The U.S. broadband industry has grown substantially over the past two decades and now touches nearly every dimension of household economic life. This paper focuses on residential broadband, which has attracted sustained federal attention and large public investments. Broadband is offered in several forms—DSL (over legacy copper telephone infrastructure), cable modem (over coaxial television networks), fiber optics (offering the highest speeds but requiring substantial upfront investment), and fixed wireless (using radio towers, practical in rural areas where wired deployment is costly)—with ISPs competing on speed, data limits, and price. Telephone and cable companies remain the dominant providers, but the industry has expanded to include national fiber entrants, fixed-wireless operators, and in some states, municipal providers.[3]

[3] Wilson [2025] finds that municipal entry crowds out private ISP investment, an important caveat for evaluating BEAD's net effect on private deployment. See Appendix A for an overview of broadband technology types.

The literature on U.S. broadband consistently returns two findings: service is expensive relative to household income, and in most local markets consumers face limited provider choice [Flamm and Varas, 2022]. In my December 2020 sample, the average census tract has 2.36 providers offering fixed broadband, and the median provider Herfindahl–Hirschman Index (HHI)—computed within tracts using provider market shares—is 5,712, well above the 2,500 threshold traditionally associated with concentrated markets. Fiber-optic coverage, which offers the best performance and reliability, is concentrated in urban and suburban areas; rural tracts disproportionately rely on DSL over aging copper infrastructure, which has lower speeds and reliability.

These market structure conditions underlie what the federal government calls the digital divide: the gap between households with reliable internet access and those without. Income, geography, and digital literacy all predict which side of that gap a household falls on, and the consequences extend to job search, healthcare, and schooling [Forman et al., 2005, Goldfarb and Prince, 2008, Goolsbee and Klenow, 2002]. The federal government has responded with a range of subsidy programs: some target affordability directly; others subsidize infrastructure deployment in areas where private investment has not materialized.

2.1.2 Federal Broadband Subsidies

Congress passed the Infrastructure Investment and Jobs Act (IIJA) in 2021, allocating roughly \$65 billion to broadband. The two main programs are the Affordable Connectivity Program (ACP), which subsidized household internet bills directly, and the Broadband Equity, Access, and Deployment (BEAD) program, which funds deployment in underserved areas. Of the total, \$14.2 billion went to ACP and \$42.45 billion to BEAD.

ACP was introduced as a successor to the Emergency Broadband Benefit (EBB). The EBB program, which ran from February to late 2021, reached nearly nine million low-income households with a subsidy of up to \$50 per month. The ACP enrolled 23 million households at its peak, offering a monthly discount of up to \$30 (\$75 on qualifying Tribal lands). Eligible households could also receive a one-time discount of up to \$100 to purchase a connected device.[4] ACP funding was exhausted in June 2024 when appropriations were not renewed by Congress. The program's termination is particularly relevant for this paper: the period 2022–2024 represents ACP's full operational lifecycle, from launch through exhaustion, and is fully covered by the reduced-form event study panel. ACP's termination also raises the immediate policy question of whether a successor program is warranted and, if so, what form it should take. The welfare and fiscal-efficiency analysis in this paper directly informs that debate.

[4] https://www.fcc.gov/acp

The BEAD program represents the largest federal investment targeting broadband providers directly. It allocates funds to states based on the number of underserved locations in each jurisdiction. The FCC defines underserved locations as broadband-serviceable locations that either lack broadband service entirely or lack access to reliable service at speeds of at least 25/3 Mbps with latency below 100 milliseconds.[5] Allocations are determined by federal formulas, so states cannot independently influence the amount of funding they receive. Although IIJA was enacted in November 2021, state-level BEAD allocations were not announced until June 2023, and states were required to submit initial proposals by December 2023. This two-year gap between enactment and allocation announcement—followed by subgrantee selection, permitting, and construction—directly explains the two-to-three year lag in market effects documented in the event study. As of 2025, states were in various stages of implementation, with deployment expected to extend through 2026 and beyond. The structural model uses a pre-IIJA baseline and provides counterfactual predictions that are best understood as the program's steady-state equilibrium effects once deployment is complete, rather than as predictions of specific intermediate-year outcomes.

2.1.3 Economic Rationale

The economic rationale for both programs rests on market failures in residential broadband. On the demand side, low income and limited digital literacy create participation barriers: even where broadband is available, many households cannot afford monthly plans. The American Community Survey reports that in 2020, roughly 15 percent of U.S. households with broadband access reported being unable to afford their current plan, and broadband subscription rates among households below 200 percent of the federal poverty line were roughly 20 percentage points below the national average. A demand-side subsidy addresses this access gap directly, lowering the effective price to a level where price-constrained households are willing to subscribe.

On the supply side, the market failure is different: fixed deployment costs are high enough to make broadband deployment privately unprofitable in low-density or low-income areas, even when the social benefits of connectivity are large. The tension between private and social returns to infrastructure investment is a classic rationale for public subsidy [Bresnahan and Reiss, 1991, Kearns, 2024]. An important caveat is that public supply-side intervention may also crowd out private entry: Wilson [2025] finds that municipal broadband provision reduces private ISP investment in affected markets, a general-equilibrium cost that standard program evaluations omit. BEAD directly addresses the deployment margin by reducing

[5] https://broadbandusa.ntia.doc.gov

the provider's private cost of entry, but the net welfare effect depends on how much private investment would have occurred in its absence.

The economic distinction matters for program design and evaluation. A demand-side subsidy like ACP has no direct effect on whether infrastructure is available in the first place; it can only stimulate adoption in markets where ISPs already offer service. In unserved areas—where infrastructure does not exist—no level of demand-side subsidy can induce adoption. BEAD, by contrast, targets the deployment decision directly. But BEAD cannot improve affordability in markets where infrastructure already exists and prices are the binding constraint. The two programs are thus economic complements: each addresses a market failure the other cannot.

An additional consideration is that the broadband market is oligopolistic, not perfectly competitive. Most U.S. households have access to at most two or three fixed-broadband providers, and many face a single dominant ISP [Flamm and Varas, 2022]. In this setting, demand-side subsidies may affect more than household take-up: a subsidy that expands the addressable market can also be associated with changes in posted prices and product offerings. This possibility motivates the reduced-form analysis, while the structural counterfactuals below keep ACP's direct demand-expansion margin analytically separate from any additional provider price adjustment.

2.2 Data Sources and Estimation Samples

The empirical analysis draws on five publicly available data sources. Each is described below with its collection period, information content, and unit of observation.

Federal Communications Commission Urban Rate Survey. The FCC collects the Urban Rate Survey (URS) annually from a stratified random sample of ISPs operating in urban census tracts, spanning 2014 to 2025.[6] Each record reports the plan's advertised download and upload speeds, technology type, monthly price, and a plan-availability weight. URS prices and speed characteristics are the primary source of product-level broadband prices in both the state-year panel and the structural estimation sample.

FCC Form 477 Deployment Data. Form 477, collected semi-annually, records where ISPs offer fixed broadband service, spanning 2014 to 2023 at the provider-census-block level.[7] I restrict to residential fixed broadband and use the December 2020 release as the structural

[6]Data available at: https://www.fcc.gov/economics-analytics/industry-analysis-division/urban-rate-survey-data-resources.

[7]Data available at: https://broadband477map.fcc.gov/#/data-download.

baseline. Form 477 records define product availability—which provider offers which speed tier in which census tract—and are the basis for constructing observed and potential products entering the entry game.

American Community Survey. The American Community Survey provides annual demographic and socioeconomic estimates at the census-tract level.[8] From the 2020 five-year release I extract housing units (market size), median household income, poverty rate, broadband subscription rate, renter share, college-educated share, and the age-65+ share; from the one-year release, the share of workers employed from home. These variables serve as market-size measures, eligibility proxies, and demand-side controls.

FCC Broadband Adoption Data. The FCC publishes tract-level fixed broadband adoption statistics derived from Form 477 subscription filings, reporting the share of households with fixed broadband connections by speed threshold in binned ranges.[9] I use the December 2020 file to anchor the product-level market share construction described in Section 2.5.

BEAD Program Allocation Records. State-level BEAD allocations, announced in June 2023, are published by the NTIA and determined by a federal formula based on certified counts of unserved broadband-serviceable locations.[10] Because states cannot influence their allocation amount, this variable serves as a predetermined continuous treatment in the event study.[11]

These five sources are combined into two estimation samples. The *state-year policy panel* covers 52 states (including the District of Columbia and Puerto Rico) annually from 2014 to 2025 and is used for the reduced-form event study. It pairs Urban Rate Survey price and plan-composition outcomes with BEAD allocation intensity and American Community Survey poverty rates. The *tract-level structural sample* is a single cross-section for December 2020 and is used for demand estimation, marginal-cost recovery, and fixed-cost bounds. It combines Form 477 product availability, Urban Rate Survey prices, FCC adoption shares, and American Community Survey demographics at the census-tract level; it contains 220,814 products in 69,085 tracts.

[8] Data available at: https://www.census.gov/programs-surveys/acs.

[9] The December 2020 release is available at: https://docs.fcc.gov/public/attachments/DOC-395959A1.pdf.

[10] State allocation totals available at: https://www.ntia.gov/funding-programs/internet-all/broadband-equity-access-and-deployment-bead-program/program-documentation/state-allocation-totals.

[11] Provider names in Form 477 contain substantial variation across subsidiaries and legacy brands. I consolidate these into canonical corporate-family identifiers; details are in Appendix C.

The state-year design is motivated by data quality as well as identification. Constructing a consistent tract-year panel over 2014–2025 is complicated by the FCC's transition from Form 477 to the Broadband Data Collection system after 2021, which altered geographic reporting units and coverage measurement. The state-year design avoids the spurious dynamics that transition would introduce. The structural model, which uses a single December 2020 cross-section, is unaffected by this transition. Appendix B reports file-level details, merge keys, and sample restrictions; the full replication package is available upon request.

2.3 Sample Construction

The structural baseline is built for December 2020, predating both the Affordable Connectivity Program and any firm-level BEAD deployment. Monthly prices for each product are assigned from the Urban Rate Survey using a six-level brand-first hierarchy—prioritizing matches by state, provider, and speed tier—that achieves a 100 % imputation rate across the 673,110 product observations. Urban Rate Survey prices are converted to constant 2025 dollars using Consumer Price Index adjustment factors. Full implementation details are provided in Appendix D.

2.4 Descriptive Statistics

Table 1 reports summary statistics for the structural estimation sample. The average broadband plan price is $71.65 per month, with substantial dispersion across providers and speed tiers. The average census tract has 2.36 providers and 3.20 observed products, consistent with the limited provider choice documented in the literature [Flamm and Varas, 2022]. Median household income is $67,152 and broadband adoption averages 66.8 percent of households. Market-share statistics are reported separately in Table 2 once the share-construction procedure is introduced in the next section.

Table 1: Descriptive Statistics: Structural Estimation Sample

Variable	Mean	Std. Dev.	P25	P75
Product-level variables				
Price ($/month)	71.65	25.34	49.99	86.17
Download speed (Gbps)	0.337	0.713	0.025	0.300
Upload speed (Gbps)	0.065	0.136	0.003	0.030
High-speed tier ($\geq$25 Mbps/$\geq$3 Mbps)	0.624	–	–	–
Tract-level variables				
Products per tract	3.20	–	2	4
Providers per tract	2.36	–	2	3
Households	1,679	820	1,126	2,089
Population	4,511	2,325	2,967	5,601
Median household income ($)	67,152	33,805	44,318	82,159
Renter share	0.364	0.228	0.183	0.509
College share	0.311	0.193	0.161	0.425
Age 65+ share	0.164	0.080	0.114	0.200
Broadband adoption share	0.668	0.171	0.559	0.803

Notes: The sample covers 220,814 tract-provider-tier products in 69,085 census tracts observed in December 2020. The high-speed tier is defined by the federal broadband standard: download speed of at least 25 Mbps and upload speed of at least 3 Mbps. Market-share statistics are in Table 2.

2.5 Market Share Construction

The central measurement challenge in estimating broadband demand is the absence of publicly available product-level market shares. The FCC reports where providers offer service, the technologies they use, and advertised speeds, but does not publicly report subscribers by provider-speed product in each census tract. This missing object is essential for my demand estimation.

I construct product-level shares by combining FCC adoption information with tract product availability and characteristics, following the logic of Crawford et al. [2019]. The procedure has two steps. First, I use FCC tract-level adoption data—reported in binned ranges by speed threshold—to recover tier-level adoption shares for each tract. Second, I allocate tier-level adoption across available products using logit weights implied by product characteristics and demographics, so that product shares sum to total fixed-broadband adoption in each tract by construction. The constructed shares are inferred quantities, not directly observed subscriber counts. *Full implementation details, the variables used in the allocation equation, and validation diagnostics are in Appendix E.*

Table 2 reports summary statistics for the constructed market shares.

Table 2: Market Share Summary Statistics

Variable	Mean	Std. Dev.	P25	P75
Product market share $\hat{s}_{jt}$	0.259	0.152	0.159	0.324
Within-tier share $\hat{s}_{j\|g,t}$	0.313	0.166	0.202	0.383
Outside-option share s_{0t}	0.171	0.134	0.100	0.300
Provider Herfindahl–Hirschman Index	5,712	2,584	3,694	6,394

Notes: Product shares are inferred using FCC adoption information and product-level data (see Section 2.5). The Herfindahl–Hirschman Index is computed within tracts using provider shares among fixed-broadband subscribers and is scaled from 0 to 10,000. The sample contains 220,814 products in 69,085 census tracts.

Table 3 benchmarks the structural estimation sample against national FCC aggregates. The sample contains 69,085 census tracts and covers 116.0 million households, or about 94 percent of the national household benchmark. The constructed adoption shares imply 92.7 million residential fixed-broadband connections at or above the 25/3 Mbps threshold, close to the corresponding FCC national benchmark. The median tract adoption ratio is also close to the FCC benchmark, supporting the use of the sample as a large-scale baseline for the structural counterfactuals. Appendix E reports the model-implied allocation of subscribers across providers as a diagnostic for the CSS share-construction procedure.

Table 3: Aggregate Validation of the Structural Sample

Moment	Structural sample	FCC national benchmark	Coverage
Households	116.02 million	123.67 million	93.82%
Residential fixed connections, $\geq$25/3 Mbps	92.75 million	94.45 million	98.20%
Median tract adoption ratio	0.90	0.89	–

Notes: The structural sample contains 69,085 census tracts. Residential fixed connections are computed as tract households multiplied by the FCC-based fixed-broadband adoption share. The $\geq$25/3 Mbps measure uses the tract-level high-speed adoption share. FCC national benchmarks are from Federal Communications Commission [2022]. Coverage is the structural-sample value divided by the FCC national benchmark. These moments validate the scale of the structural sample against national residential broadband aggregates; they are not provider-level subscriber counts.

Broadband market structure. The market structure of the residential broadband industry shapes both the identification strategy and the interpretation of the structural results. Table 4 summarizes the distribution of provider counts in the estimation sample.

Table 4: Provider Competition in the Sample

Number of providers in tract	Share of tracts (%)	Share of households (%)
One provider	23.10	21.96
Two providers	40.16	39.22
Three or more providers	36.73	38.82

Notes: Provider counts based on canonical providers with positive constructed product shares in the tract. Sample: 69,085 census tracts in December 2020.

Nearly a quarter of census tracts in the sample have a single fixed-broadband provider, and almost two-thirds have at most two. This concentration is one of the market failures the federal programs address. Monopoly tracts account for 21.96 percent of households—a substantial fraction facing no intra-market price competition for fixed broadband. The moderate share of tracts with three or more providers (36.73 percent) reflects that competitive broadband markets, while they exist, are not the norm.

3 Reduced-Form Evidence

3.1 State-Year Panel and Treatment Variables

The reduced-form analysis uses a balanced panel of 52 U.S. states (including Puerto Rico and the District of Columbia) observed annually from 2014 to 2025, yielding $N = 624$ state-year observations before missing values. The outcomes are drawn from the FCC Urban Rate Survey (URS), which records plan-level prices, speeds, and technology types at the state level each year. I aggregate to a state $\times$ year panel and compute weighted-average prices (weights equal the number of plans) and plan composition shares. Real prices are deflated to 2025 USD using the Consumer Price Index for Urban Consumers (CPI-U). I examine six outcomes organized into three groups: prices (log real weighted-average price and log prices by speed tier), plan composition (shares of plans below and above the 25/3 Mbps threshold), and market structure (provider Herfindahl–Hirschman Index). Table 5 reports descriptive statistics.

Table 5: Descriptive Statistics – Main Outcomes

Outcome	N	Mean	SD	Min	Median	Max
Prices						
Log avg. real price	597	4.583	0.190	3.827	4.556	5.254
Log price (<25/3 Mbps)	563	4.200	0.210	3.314	4.223	5.073
Log price (≥25/3 Mbps)	587	4.703	0.223	4.169	4.673	5.483
Plan Composition						
Plan share (<25/3 Mbps)	597	0.261	0.260	0.000	0.171	1.000
Plan share (≥25/3 Mbps)	597	0.739	0.260	0.000	0.829	1.000
Market Structure						
Provider Herfindahl–Hirschman Index	597	0.524	0.238	0.146	0.477	1.000

Notes: Unit of observation is state × year, 2014–2025 ($N = 624$ state-year cells before missing). These six outcomes are the main outcomes in the event studies and difference-in-differences regressions. Prices are weighted-average plan prices deflated to 2025 USD using the Consumer Price Index for Urban Consumers. Plan shares and the Herfindahl–Hirschman Index are bounded $[0, 1]$.

Panel structure and outcome construction. The state-year panel spans 2014–2025, providing pre-ACP years (2014–2021), three ACP years (2022–2024), and one follow-up year (2025). For BEAD, the panel provides six pre-announcement years (2014–2020), one year following IIJA enactment (2021), and four post-enactment years (2022–2025). This timing is useful for the event study design: five pre-period observations provide enough pre-trend tests to have power against violations of parallel trends, while four to five post-period observations are sufficient to identify differential dynamic effects.

The Urban Rate Survey collects plan-level data from a stratified sample of urban census tracts annually. Each record reports a plan's advertised download and upload speeds, monthly charges, and technology type. I compute state-year weighted averages and technology shares from these plan-level records. Because Urban Rate Survey sampling targets urban areas, the state-year panel is best interpreted as capturing urban and suburban broadband market conditions. Rural broadband dynamics, which are the primary target of BEAD, are underrepresented in the Urban Rate Survey-based outcomes. This means the event study estimates of BEAD effects are likely conservative: if BEAD's price and concentration effects are disproportionately concentrated in rural areas (as the structural analysis suggests), Urban Rate Survey-based outcomes will understate the aggregate BEAD impact. In contrast, the Affordable Connectivity Program's effects, which operate through demand expansion in served (predominantly urban and suburban) markets, are more directly captured by the Urban Rate Survey sampling frame.

Treatment intensity variables are both time-invariant and predetermined. This design avoids the endogenous variation concerns that arise when treatment intensity is measured contemporaneously with outcomes. The Affordable Connectivity Program eligibility share (FPL200_i) is drawn from the 2020 American Community Survey, before the program was implemented. The BEAD intensity variable ($\log \text{BEAD}_i$) is determined by federal allocation formulas based on FCC-certified unserved location counts, which are also measured before BEAD grants were disbursed. Neither measure can be influenced by ISP behavior or state broadband outcomes over the event window, supporting the identifying assumption that differential treatment assignment is uncorrelated with differential trends.

ACP exposure. I measure state-level ACP eligibility exposure as the share of households with incomes below 200% of the federal poverty level (FPL200_i), drawn from the 2020 American Community Survey. This is a time-invariant, pre-determined cross-sectional measure that varies across states and proxies for the potential demand stimulus ACP can generate. The program became active in 2022; I set $\text{Post}_t^{\text{ACP}} = \mathbf{1}[t \geq 2022]$.

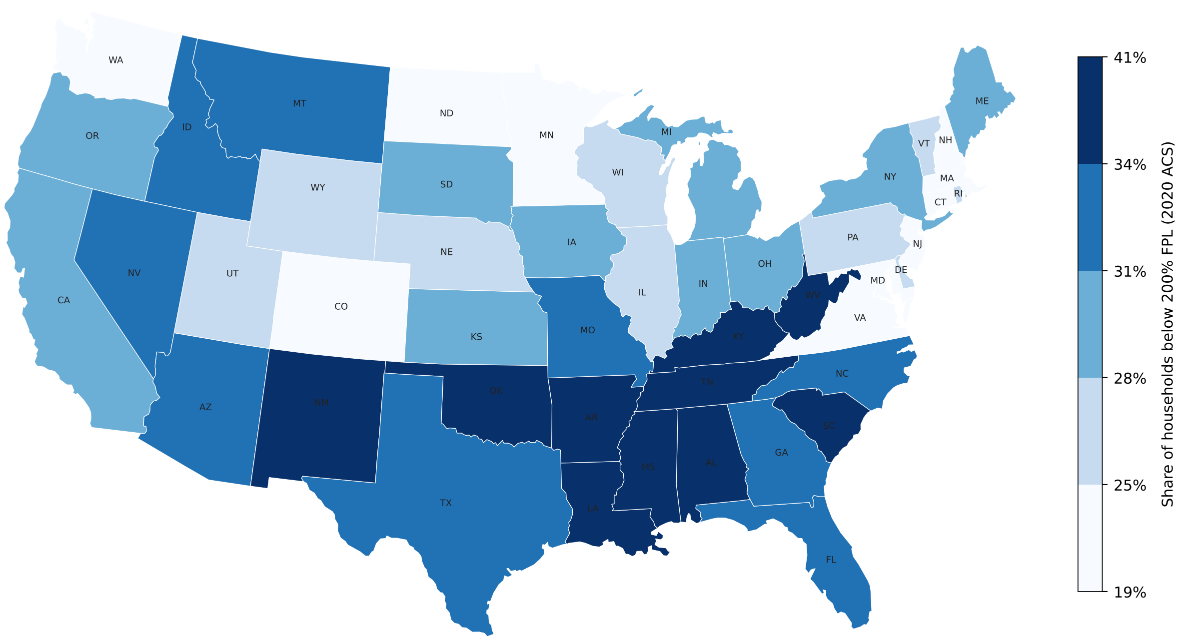

Figure 1: State-level ACP exposure: share of households with incomes below 200% of the federal poverty level (FPL200_i), measured from the 2020 American Community Survey. Darker shading indicates higher eligibility exposure. Colour breaks at sample quintiles.

BEAD exposure. I measure state-level BEAD intensity as the log of BEAD allocation (USD) per unserved broadband location ($\log \text{BEAD}_i$). A higher value indicates that a state receives more funding per targeted location, implying a more intensive supply-side intervention. This measure is also time-invariant and pre-determined by federal allocation formulas; states cannot influence their allocation. BEAD was announced in 2023; I set $\text{Post}_t^{\text{BEAD}} = \mathbf{1}[t \geq 2023]$.

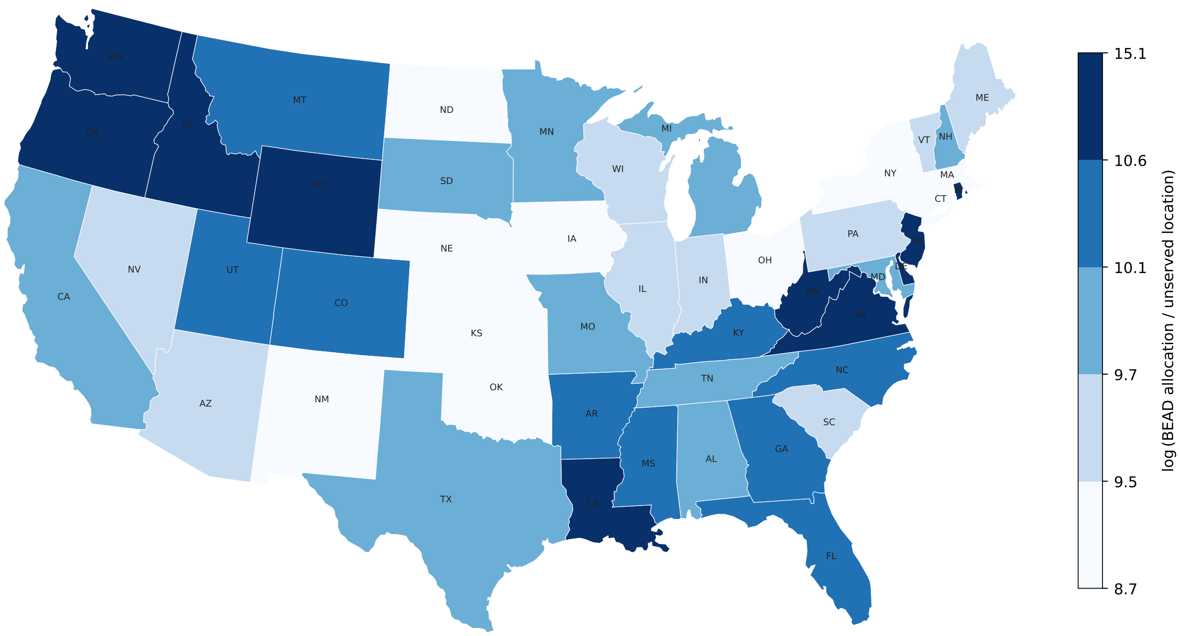

Figure 2: State-level BEAD exposure: log of BEAD allocation (USD) per unserved broadband location ($\log \text{BEAD}_i$), based on NTIA allocation announcements (June 2023). Darker shading indicates higher funding intensity per targeted location. Colour breaks at sample quintiles.

3.2 Identification Strategy

For program $P \in \{\text{ACP}, \text{BEAD}\}$, let D_i^P denote the time-invariant treatment intensity for state i and let τ index event time relative to the policy year. The dynamic specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^{\bar{\tau}^P} \beta_\tau^P \Big(D_i^P \times \mathbf{1}[t = \bar{t}^P + \tau] \Big) + \boldsymbol{\gamma}_i' \, t + \varepsilon_{it}, \tag{3.1}$$

where α_i are state fixed effects, λ_t are year fixed effects, $\bar{t}^{\text{ACP}} = 2022$ and $\bar{t}^{\text{BEAD}} = 2021$. The omitted period is $\tau = -1$ (the year before the policy), so all estimates are relative to that baseline. Regressions are weighted by the number of plans n_{it} and standard errors are clustered at the state level.

The role of fixed effects. State fixed effects α_i absorb all time-invariant determinants of broadband outcomes at the state level, including permanent differences in geographic coverage, regulatory environment, and demographic composition. Identification therefore exploits *within-state* variation in outcomes over time, purged of any level differences across states. Year fixed effects λ_t absorb common national shocks to broadband markets, such as economy-wide shifts in input costs, federal regulatory changes, or aggregate demand trends. The interaction of treatment intensity with year indicators then captures whether states more intensively exposed to each program moved differentially relative to the national trend.

State-specific trends. For both ACP and BEAD, I include state-specific linear time trends $\boldsymbol{\gamma}_i' t$ to account for the possibility that treatment intensity correlates with pre-existing state-level broadband trajectories. For ACP, states with higher poverty rates—which mechanically have higher eligibility exposure—may have been following divergent broadband price and adoption paths before the program launched. For BEAD, states with more unserved locations tend to be lower-density and lower-income, and may have been experiencing systematically different infrastructure investment trends prior to IIJA enactment. The digital divide literature has documented that broadband adoption and pricing trends differ systematically across income and demographic groups [Goldfarb and Prince, 2008, Espín and Rojas, 2024]. Without state-specific trends, such pre-existing divergence would contaminate both sets of estimates. State-specific trends absorb smooth state-level deviations from the national year effects; the treatment coefficients $\hat{\beta}_\tau^{\text{ACP}}$ and $\hat{\beta}_\tau^{\text{BEAD}}$ are then identified from non-linear departures from each state's pre-existing trend coinciding with program rollout.

Assumption 1 (Conditional parallel trends)**.** *Conditional on state fixed effects, year fixed effects, and state-specific linear time trends, the untreated potential outcome follows the same trend across states with different treatment intensities D_i^P. Formally, $\mathbb{E}[\varepsilon_{it} \mid D_i^P, i, t] = 0$ for all $\tau < 0$.*

The identifying assumption is that states with higher treatment intensity D_i^P were not on systematically different trajectories for reasons unrelated to the policy. Because both FPL200_i and $\log \text{BEAD}_i$ are predetermined by pre-program demographic composition and federal allocation formulas, the main threat to identification would be that states with higher exposure were on different broadband trends before the programs launched. The pre-period coefficients $\hat{\beta}_\tau^P$ for $\tau < 0$ serve as direct falsification tests of Assumption 1.

A secondary concern is the anticipation of program benefits: ISPs or households might alter behavior before formal program launch in expectation of the policy. Under the IIJA timeline, BEAD was announced in November 2021 but allocations were not finalized until 2023, so anticipatory responses would appear as effects at $\tau = 1$–2 (2022–2023), which is largely consistent with what I find (early effects are near zero). For the Affordable Connectivity Program, its predecessor (the Emergency Broadband Benefit program) ran from February to late 2021, meaning some anticipatory ISP behavior may already be absorbed in the $\tau = -1$ base period; this would lead me to understate ACP's total effect. Neither concern appears to drive the main qualitative results.

3.3 ACP: Event Study Results

Figure 3 presents event study estimates from (3.1) for six key outcomes. Event time is indexed relative to 2022 ($\tau = 0$); the dashed vertical line marks November 2021 (IIJA enactment). The omitted period is $\tau = -1$ (year 2021).

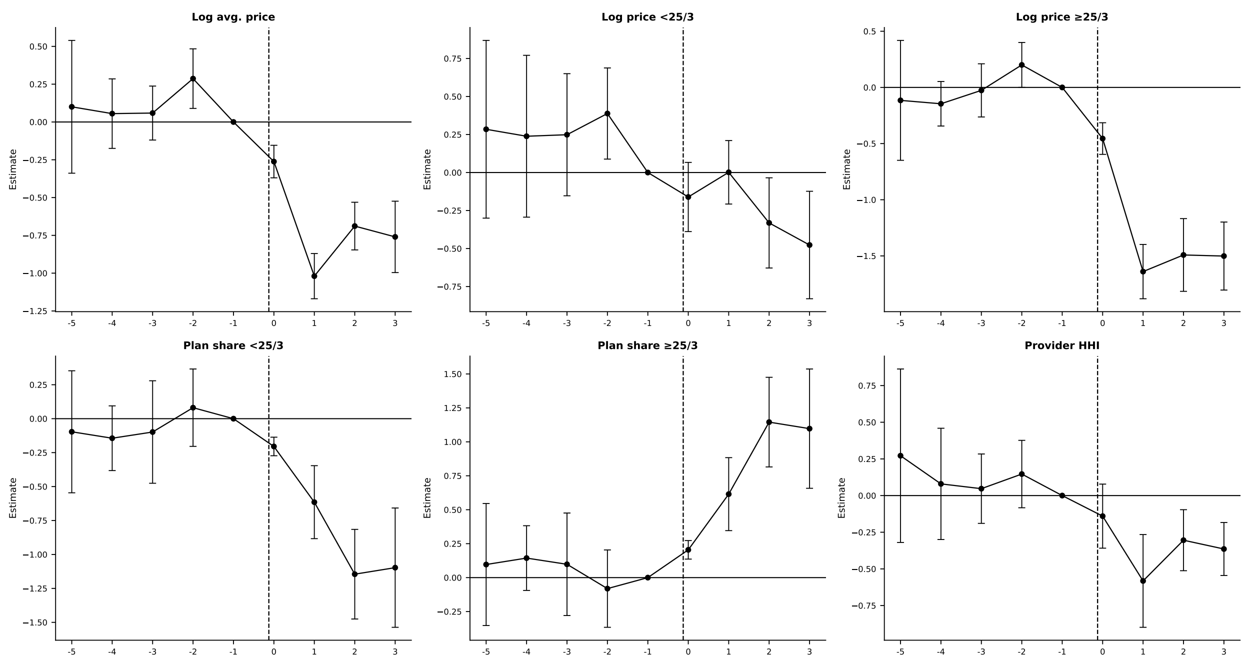

Figure 3: ACP event study estimates. Treatment intensity: share of households below 200% FPL. TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Bars indicate 95% confidence intervals.

Pre-trends. The pre-period pattern is broadly consistent with the parallel-trends assumption, conditional on state and year fixed effects and state-specific linear trends. For most outcomes (log prices for high-speed plans, plan shares, and provider HHI), the pre-period coefficients are small and close to zero. Log price for plans below 25/3 Mbps shows somewhat elevated pre-period point estimates at $\tau = -5$ to $\tau = -3$, though with wide confidence intervals. This motivates the inclusion of state-specific linear trends, which absorb differential pre-existing trajectories in broadband prices across states at different poverty rates.

Post-ACP effects. A clear pattern emerges in the post-period. Average broadband prices fall significantly in higher-eligibility states following ACP's 2022 launch. The coefficient $\hat{\beta}_1^{\text{ACP}}$ for log average price is approximately -1.0 per unit of FPL200 share. To translate this into economically interpretable magnitudes, the interquartile range of state FPL200 shares in my panel is approximately 0.10 (from about 0.27 at the 25th percentile to 0.37 at the 75th). Moving from the 25th to the 75th percentile of ACP eligibility exposure is therefore associated with an additional log average price decline of roughly $-0.10 \times 1.0 \approx 10$ log points (approximately 10 percent) at peak ($\tau = 1$), stabilizing at roughly -7 percent through $\tau = 2$–3.

The price decline is especially pronounced for high-speed plans (log price $\geq$25/3 Mbps), where the estimated response is approximately -0.15 to -0.18 log points per percentage-point increase in FPL200 share at $\tau = 1$, implying a 75th-to-25th percentile comparison of roughly 15–18 percent. This effect persists through $\tau = 2$–3. Plan composition shifts substantially: the share of low-speed plans (below 25/3 Mbps) contracts through $\tau = 2$–3, while the high-speed share expands commensurately, consistent with ISPs upgrading their product portfolios in response to elevated demand for faster service from newly subsidized households. Provider HHI declines in higher-exposure states, though the estimates are noisy and should be interpreted with caution.

The reduced-form evidence therefore points to more than a mechanical increase in subsidized subscriptions. In high-eligibility states, observed prices fell and the mix of plans shifted toward higher-speed offerings. I interpret this as evidence that ACP expanded the relevant customer base for ISPs, while leaving the exact decomposition of the price movement—pass-through, strategic pricing, or changes in plan composition—to the structural and institutional discussion below.

3.4 BEAD: Event Study Results

Figure 4 presents event study estimates for BEAD. Event time is indexed relative to 2021 (IIJA enactment, $\tau = 0$); the omitted period is $\tau = -1$ (year 2020). The specification follows (3.1): state and year fixed effects and state-specific linear time trends $\boldsymbol{\gamma}_i' t$, weighted by number of plans, with standard errors clustered at the state level. State-specific trends are warranted for BEAD because states with more unserved locations tend to be lower-density and lower-income and may have been following systematically different infrastructure investment trajectories prior to IIJA enactment; the trends absorb these smooth pre-existing differentials so that the BEAD coefficients are identified from non-linear departures coinciding with program rollout.

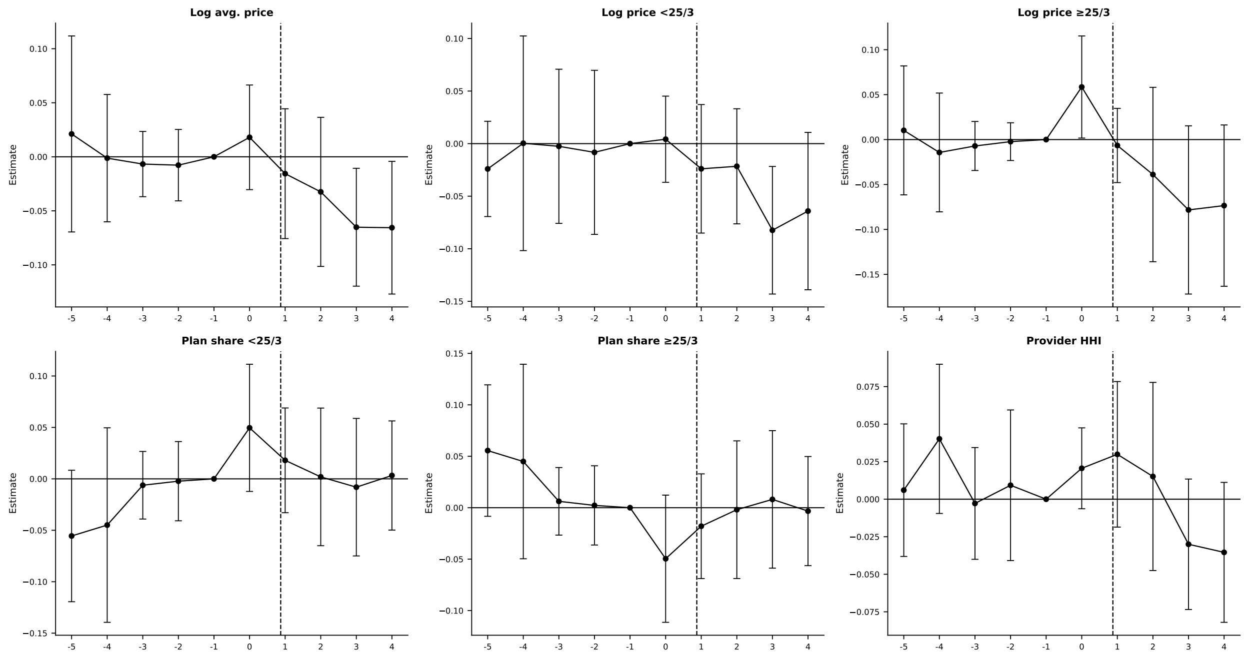

Figure 4: BEAD event study estimates. Treatment intensity: log BEAD allocation per unserved location. TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2021 (IIJA enactment); the dashed vertical line marks $\tau = 1$ (2022), the first year following enactment. Bars indicate 95% confidence intervals.

Pre-trends. Pre-period coefficients are small and statistically indistinguishable from zero across all outcomes. States that ultimately received more BEAD funding per unserved location were on similar trajectories in prices, technology, and market structure before the program was announced.

Post-BEAD effects. BEAD produces smaller effects than ACP, and they materialize with a longer lag. In the first two years following IIJA enactment ($\tau = 1$–2, 2022–2023), effects on all outcomes are minimal and statistically indistinguishable from zero. Price responses become more apparent at $\tau = 3$–4 (2024–2025), with log average price declining by roughly 0.05–0.07 log points per log unit of BEAD intensity. Provider HHI shows a similarly lagged response, turning negative at $\tau = 3$–4 after being near zero or slightly positive in the first two post-enactment years. Plan composition, by contrast, shows no discernible shift over the event window: the shares of high-speed and low-speed plans are statistically indistinguishable from zero throughout $\tau = 1$–4. This timing fits the institutional rollout of BEAD. The event window captures only the early phase of infrastructure deployment: new networks must be built, interconnected, and activated before entry can discipline incumbent pricing.

3.5 Summary Difference-in-Differences Estimates

To complement the event study evidence, Table F.1 (Appendix F) reports static two-way fixed-effects difference-in-differences estimates for each outcome, pooling the post-period. These estimates summarize the average treatment effect over the post-program period and are directly comparable across outcomes and programs. They serve as a robustness check on the event study patterns: if the dynamic patterns are genuine, the static DiD should be significant and in the same direction as the post-period event study coefficients.

The DiD estimates tell the same story. For ACP, prices fall significantly in higher-eligibility states (log average price coefficient -0.503, log high-speed price -0.990), fiber share rises ($+0.347$), DSL share falls (-0.442), and provider HHI declines (-0.148). These estimates are consistent with price and product-mix adjustments in higher-exposure states. For BEAD, estimates are smaller and less precisely estimated, consistent with the event study showing effects concentrated in the later post-enactment years. Provider HHI and top-provider share show modest negative effects, consistent with the lagged entry response documented in the event study. The absence of significant plan-composition effects for BEAD confirms that BEAD operates primarily through the entry margin rather than product repositioning.

3.6 What the Reduced-Form Evidence Establishes

The event studies document the direction, timing, and reduced-form magnitude of each program's effects, but three limitations define the scope of the structural analysis that follows. First, the observed price declines are consistent with multiple channels—direct pass-through, strategic pricing, or changes in plan composition—while the structural model quantifies demand expansion while holding gross provider prices fixed; the two exercises therefore answer different questions. Second, state-level estimates capture average effects across heterogeneous local markets, whereas the structural model exploits rich tract-level variation in market structure, provider presence, and cost conditions that the state panel cannot capture. Third, the event study cannot evaluate alternative policy designs, decompose welfare into consumer and producer surplus, or identify the entry and product-choice responses driving the observed price and concentration patterns. Standard evaluations that count new subscribers and dollars per connection miss these equilibrium responses [Goolsbee and

Guryan, 2006]; the structural model makes them central.[12] The reduced-form and structural analyses are therefore complements: the former documents that the programs changed market outcomes; the latter quantifies the channels the model is built to capture and enables counterfactual policy comparisons.

4 Structural Model

4.1 Model Setup and Timing

Markets and products. Markets are indexed by t and firms (or providers) by f. A *market* is a U.S. census tract in 2020, before the implementation of the major IIJA broadband programs.[13] A *product position* is a provider-speed-tier pair (f, g, t), where $g \in \{H, L\}$ denotes the IIJA high-speed tier ($\geq$25/3 Mbps) and low-speed tier (<25/3 Mbps). This classification captures a coarse but economically meaningful distinction in service quality relevant for household adoption decisions, distinguishing connections that can reliably support high-bandwidth activities from more limited service options. The indicator Y_{fgt} equals one if provider f offers tier g in tract t and zero otherwise; $Y_t = \{Y_{fgt}\}_{(f,g)\in\mathcal{J}_t^0}$ summarizes the product configuration. Provider entry in this model is *product positioning*: a provider that already serves some tracts in a county expands its product line by activating a tier in an additional tract or offering a second speed tier in an existing tract. The model thus captures both geographic expansion and product-line decisions within the local market.

The *outside option* is no fixed-broadband subscription. Market size M_t is the number of households in the tract, and the market share of the outside option is $s_{0t} = 1 - \sum_{j\in\mathcal{J}_t} s_{jt}$.

Timing and information. The following two assumptions formalize the timing and the information structure.[14]

Assumption 2 (Timing). *The game proceeds in two stages:*

[12] Because ACP was introduced nationally at a common date and exposure varies only through time-invariant eligibility shares, the TWFE estimator is not subject to staggered-adoption bias [Sun and Abraham, 2021, Goldsmith-Pinkham et al., 2020]; the coefficients trace differential dynamic responses to continuous treatment intensity. Appendix F reports static DiD estimates across additional specifications; the qualitative conclusions are stable.

[13] I use census-tract data and pre-2021 FCC Form 477 deployment data to ensure the baseline predates IIJA programs. This timing is important for the counterfactuals: simulated price discounts and fixed-cost reductions are applied to markets that do not already reflect ACP or BEAD effects.

[14] The timing assumption has been used in several studies. Examples include Fan and Yang [2024] for the brewery industry, Eizenberg [2014] in the personal computer industry, and Hidalgo [2024] for the broadband industry.

1. *Product-positioning stage. Firms observe demand shifters X_t, cost shifters W_t, rivals' observed footprints, and their own private fixed-cost shocks ζ_{fgt}. Based on this information, each firm simultaneously chooses its portfolio $Y_{ft} \in \mathcal{Y}_{ft}$.*

2. *Pricing stage. Conditional on the realized product configuration Y_t, firms observe demand shocks ξ_{jt} and marginal-cost shocks ω_{jt} and simultaneously set monthly plan prices in a complete-information Bertrand-Nash game.*

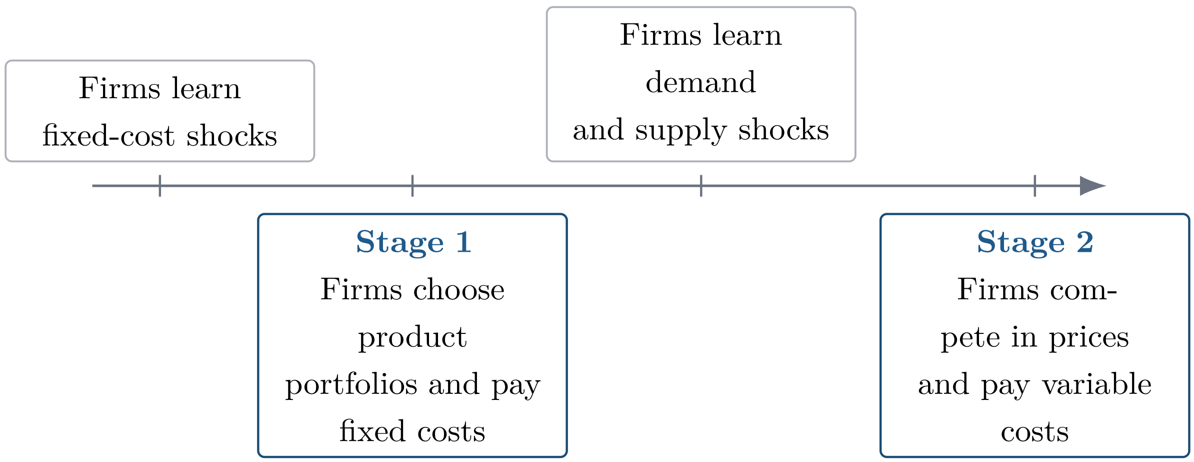

Figure 5: Timing of the structural entry and pricing game.

Assumption 3 (Information)**.** *Firms. At the product-positioning stage, fixed-cost shocks ζ_{fgt}, market characteristics (X_t, W_t), and rivals' potential market footprints are all common knowledge. The demand and cost shocks (ξ, ω) that will be realized in the second-stage pricing game are not yet known; only their joint distribution is common knowledge. Firms therefore compute expected variable profits by integrating over (ξ, ω).*

Econometrician. The shocks (ξ, ω, ζ) are unobserved. Market characteristics (X_t, W_t) and the set of observed product choices $\{Y_{fgt}\}$ are observed.

The separation between portfolio choice and pricing is standard in the entry and product-positioning literature [Mazzeo, 2002, Seim, 2006, Fan, 2013] and reflects the different time horizons involved: infrastructure deployment decisions are made months or years before price lists are set. The complete-information assumption on fixed costs is both analytically convenient and economically natural, as ISP deployment costs depend on local permitting, right-of-way negotiations, and contractor availability that are broadly understood within local infrastructure markets. Likewise, price equilibria are modeled under complete information because prices are publicly advertised and routinely monitored by competing firms.

Firm f's first-stage objective is

$$\max_{Y_{ft} \in \mathcal{Y}_{ft}} \mathbb{E}_{\xi,\omega} \left[r_t^f \left(Y_{ft}, Y_{-ft} \right) \mid I_{ft} \right] - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}, \tag{4.1}$$

where $Y_{ft} = (Y_{fHt}, Y_{fLt})$, Y_{-ft} denotes rivals' portfolios, and $r_t^f(\cdot)$ is variable profit from the pricing game. Because a provider can offer at most two tiers in a tract, $\mathcal{Y}_{ft}$ contains four portfolios: no product, high-speed only, low-speed only, and both tiers.

The empirical difficulty is that Y_t is high dimensional. With roughly ten potential provider-tier positions in the average tract, a full solution of the entry game would require enumerating hundreds or thousands of product configurations per market. I therefore estimate first-stage fixed costs using the scalable discrete-game method of Fan and Yang [2024]. The key idea is to avoid solving the game during estimation. Instead, the model uses bounds on the incremental profit from offering each product position, which imply moment inequalities for the fixed-cost parameters. Once fixed costs are disciplined by these inequalities, counterfactual policies are evaluated by solving a best-response game under the new policy environment.

4.2 Consumer Demand

Household h's indirect utility from product j in market t is

$$U_{hjt} = \delta_{jt} + \nu_{hg} + (1-\rho)\varepsilon_{hjt}, \tag{4.2}$$

where δ_{jt} is mean utility, ν_{hg} is a group-level taste shock, and ε_{hjt} is Type-I extreme value.[15] Mean utility is

$$\delta_{jt} = X'_{jt}\beta + \alpha p_{jt} + \xi_{jt}. \tag{4.3}$$

The vector X_{jt} includes download speed, upload speed, their squares, and a high-speed tier indicator. The parameter α is the marginal utility of price and ξ_{jt} is unobserved quality.

I use a single inside nest: all fixed-broadband products compete against the outside option of no fixed-broadband subscription. The single-nest specification is chosen for two reasons. First, it maintains a consistent nesting structure across all 69,085 census tracts. A two-level nest based on speed tier (e.g., high-speed products in one nest, low-speed in another) would

[15] This paper departs from Fan and Yang [2024], who employ a random-coefficients demand model. Although such models allow for richer substitution patterns, they may generate non-unique pricing equilibria in multiproduct settings [Nocke and Schutz, 2018], thus complicating welfare and entry counterfactuals. Therefore, I use the nested logit model, which delivers a unique closed-form pricing solution and ensures a stable mapping from demand primitives to prices (e.g., Berry [1994]). Tractability is essential for evaluating firms' endogenous product offerings and entry decisions. I nonetheless rely on Fan and Yang [2024]'s framework to recover fixed-cost parameters in the entry stage.

be degenerate in markets where only one tier is offered: the within-nest share equals one, and the second-level nest collapses to a standard logit, producing an inconsistent likelihood across markets. Because a substantial fraction of rural tracts in the sample have only low-speed products, a tier-based two-level nest is not feasible without restricting the sample or treating single-tier markets separately. Second, the single-nest structure delivers a unique closed-form Bertrand-Nash pricing solution, which is essential for the entry counterfactuals: at each iteration of the product-positioning game, equilibrium prices must be computed rapidly across tens of thousands of markets and fixed-cost draws. A richer nesting structure (random coefficients, two-level nest) may admit multiple pricing equilibria in multiproduct settings [Nocke and Schutz, 2018], complicating the entry-game solution. The single-nest model therefore reflects a deliberate tractability choice, not a belief that the two-level structure is substantively incorrect. The Berry inversion [Berry, 1994] gives

$$\log(s_{jt}) - \log(s_{0t}) = X_{jt}'\beta + \alpha p_{jt} + \rho \log(s_{j|g,t}) + \xi_{jt}, \tag{4.4}$$

where $s_{j|g,t} = s_{jt}/(1 - s_{0t})$. In the tract-level data, the left-hand side is the CSS mean utility recovered from product-level implied shares.

Price and the within-inside share are endogenous. I instrument for these variables using BLP-style rivals' characteristics [Berry et al., 1995]:

$$z_{jt}^{(k)} = \frac{1}{J_t - 1} \sum_{\ell \neq j} X_{\ell t}^{(k)}, \qquad k \in \{\text{downstream}, \text{upstream}, \text{Tier H}\}, \tag{4.5}$$

and the number of same-tier rivals [Aguirregabiria et al., 2024].

The logic of these instruments is standard [Berry et al., 1995, Nevo, 2001, Conlon and Gortmaker, 2020]. Rivals' speed characteristics affect a firm's own price only through strategic interaction: a firm with slower competitors can charge higher prices without losing as many subscribers. These characteristics are plausibly exogenous to a given firm's unobserved quality shock ξ_{jt} if firms choose deployment locations for reasons uncorrelated with rivals' cost shocks. The number of same-tier rivals captures the direct competitive pressure within the speed tier, which affects pricing but is not itself a choice variable of the focal firm in any given tract.

The preferred specification absorbs State and Provider $\times$ Tier fixed effects. State fixed effects absorb time-invariant state-level quality differences in broadband markets. Provider $\times$ Tier fixed effects absorb all provider–tier–specific mean quality differences, eliminating any systematic differences between, say, fiber products versus DSL products or between large national ISPs and small regional ones. Identification of $\hat{\alpha}$ and $\hat{\rho}$ in this specification comes from within-provider-tier variation across tracts in relative prices and market shares, conditional on state-level and provider-tier-level averages.

4.3 Oligopoly Price Competition

Assume that firm f's variable profit in market t is

$$r_t^f = \sum_{j \in \mathcal{J}_t^f} (p_{jt} - mc_{jt}) M_t s_j(p_t, X_t, \xi_t), \tag{4.6}$$

where M_t is market size. The Bertrand-Nash first-order condition for product j is

$$s_{jt} + \sum_{k \in \mathcal{J}_t^f} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0. \tag{4.7}$$

Stacking conditions across products yields

$$\hat{\mu}^{BN} \equiv p - \hat{mc} = -H^{-1} s, \quad \text{where } H_{jk} = (\partial s_k / \partial p_j) * \Omega_{jr} \tag{4.8}$$

$$\Omega_{jr} = \begin{cases} 1 & \text{if } \exists f : r, j \in \mathcal{J}_i^f \\ 0 & \text{otherwise,} \end{cases}$$

I recover markups and marginal costs tract by tract and then estimate

$$\log(mc_{jt}) = W_{jt}' \gamma + \omega_{jt}, \tag{4.9}$$

where W_{jt} contains speed characteristics and the high-speed tier indicator.

4.4 Product Positioning and Fixed Costs

The supply-side object is a product-positioning decision. A product position is a provider $\times$ speed-tier pair in a census tract. Let f index canonical providers, $g \in \{H, L\}$ index speed tiers, and t index census tracts. The primitive decision is

$$Y_{fgt} = \mathbf{1}\{\text{provider } f \text{ offers speed tier } g \text{ in tract } t\}. \tag{4.10}$$

This definition treats entry as product positioning rather than entry into the national broadband industry. It captures both geographic expansion into a tract and product-line expansion within a tract. For example, a provider that already offers the low-speed tier in tract t enters the high-speed position if it also begins offering the high-speed tier.

Potential product positions. The observed data are organized at the tract × canonical-provider × speed-tier level, so each observed row corresponds to $Y_{fgt} = 1$. To estimate fixed costs, I also need feasible positions that are not chosen. I define the potential set using local provider footprints rather than assuming every national provider can enter every tract. Let $c(t)$ denote the county containing tract t. The baseline potential set is

$$\mathcal{J}_t^0 = \{(f, g) : \exists\, t' \in c(t) \text{ such that } Y_{fgt'} = 1\}\,. \tag{4.11}$$

Thus, provider-tier position (f, g) is feasible in tract t if that same provider-tier pair is observed somewhere else in the county. This restriction is economically natural: broadband expansion depends on nearby network presence, local permitting, and rights of way. It also keeps the potential choice set local enough to avoid treating providers with no plausible footprint as potential entrants.

For every feasible but unobserved position in $\mathcal{J}_t^0$, I impute product characteristics using predetermined information: provider-tier-state means, then provider-tier national means, then tier means. Observed positions retain their observed characteristics.

Table 6 reports the implied dimensions. The average tract contains 3.20 observed products but 10.54 potential provider-tier positions. A full market-level portfolio model would require evaluating $2^{|\mathcal{J}_t^0|}$ configurations per market (up to thousands at the 75th percentile), motivating the Fan and Yang [2024] approach below.

Table 6: Dimension of Observed and Potential Product Sets

Product-set definition	Mean	P25	Median	P75	P90
Observed products in tract, J_t^{obs}	3.20	2	3	4	5
Observed providers in tract	2.36	2	2	3	4
County-footprint positions, $\lvert\mathcal{J}_t^0\rvert$	10.54	6	9	13	21
County providers	8.13	4	7	10	16

Notes: The unit of observation is a census tract. Observed products are provider-tier products present in the tract. The county-footprint potential set includes provider-tier positions observed somewhere in the same county as the tract.

4.5 Multiple Equilibria and Equilibrium Selection

The product-positioning game generically admits multiple Nash equilibria: for a given draw of fixed costs, there may be several product configurations that are mutually best-response stable. This is a fundamental challenge in empirical models of entry, recognized since Bresnahan and Reiss [1991] and Tamer [2003], and addressed directly by Ciliberto and Tamer [2009] and Fan and Yang [2024].

The estimation stage does not require equilibrium selection because the moment-inequality approach only requires that the observed product configuration is *not dominated*. Every Nash equilibrium satisfies this condition, so the bounds on fixed costs are valid regardless of how equilibrium is selected in practice. This is precisely the advantage of the Fan and Yang [2024] method over likelihood-based approaches that require a unique equilibrium assumption.

The counterfactual stage does require solving for post-policy equilibria. I handle this as follows. For each market and each draw of fixed costs, I run iterated best-response from three starting points: the observed pre-policy configuration, the empty configuration, and the full feasible configuration.[16] The algorithm is declared converged when no firm can profitably deviate from the current portfolio, checked by a firm-by-firm Nash verification. If multiple fixed points are found from different starting configurations, I retain all of them. The reported national-level bounds combine equilibrium multiplicity and fixed-cost parameter uncertainty by stacking market-level lower and upper outcomes across equilibria and fixed-cost draws and reporting the 2.5th and 97.5th percentile bounds.

Markets where no fixed point is found within the iteration limit (fewer than one percent of markets) are assigned the pre-policy configuration. This conservative fallback ensures the bounds remain valid; it is likely to understate the response of hard-to-solve markets. Extensive convergence diagnostics, reported in Appendix J, show that the algorithm converges in the vast majority of markets, and the distribution of convergence failures is not systematically correlated with market size or policy intensity.

The main measurement assumptions underlying the structural analysis—inferred product-level market shares, imputed URS prices, and a static December 2020 baseline—are discussed in Appendix G, along with supporting robustness checks.

5 Estimation Results

I estimate the model in three steps. First, I estimate the nested-logit demand system using the Berry inversion in (4.4). The preferred specification instruments for price and the within-inside share using rivals' product characteristics and the number of same-tier rivals, and absorbs State and Provider $\times$ Tier fixed effects. Second, I use the demand estimates and the Bertrand-Nash first-order conditions in (4.7) to recover markups and marginal costs tract by tract. Third, I estimate fixed costs from the product-positioning inequalities implied by the first-stage model. The moment inequalities follow Fan and Yang [2024]: observed product choices are required not to be dominated, and the resulting fixed-cost parameters are reported as bounds. Appendix J gives the step-by-step construction of the fixed-cost moments, the

[16] Fan and Yang [2024] consider two starting points: the empty and the full feasible configuration.

weight matrix, and the confidence region.

5.1 Consumer Demand

Table 7 reports four demand specifications. Columns (1) and (2) are OLS; columns (3) and (4) instrument for price and the within-inside share. Even-numbered columns absorb State and Provider × Tier fixed effects. Column (4) is the preferred specification.

Table 7: Nested Logit Demand Estimates

	OLS		2SLS (BLP instruments)	
	(1)	(2)	(3)	(4)
Price / \$100 [$\hat{\alpha}$]	-0.260^{***}	-0.329^{***}	0.818^{***}	-9.424^{***}
	(0.002)	(0.004)	(0.005)	(0.696)
Within-share [$\hat{\rho}$]	0.245^{***}	0.214^{***}	0.255^{***}	0.220^{***}
	(0.001)	(0.001)	(0.002)	(0.008)
Download speed (Gbps)	0.308^{***}	0.367^{***}	0.139^{***}	0.435^{***}
	(0.003)	(0.003)	(0.010)	(0.015)
Download speed^2	-0.098^{***}	-0.107^{***}	-0.079^{***}	-0.152^{***}
	(0.002)	(0.002)	(0.009)	(0.004)
Upload speed (Gbps)	0.008^{***}	0.017^{***}	0.248^{***}	0.021^{*}
	(0.003)	(0.003)	(0.009)	(0.011)
Upload speed^2	0.047^{***}	0.051^{***}	0.022^{**}	0.089^{***}
	(0.002)	(0.002)	(0.009)	(0.003)
High-speed indicator	0.122^{***}	n.s.	-0.088^{***}	n.s.
	(0.001)	—	(0.002)	—
N	220,821	220,821	220,821	220,821
R^2	0.782	—	—	—
State FEs	No	Yes	No	Yes
Provider×Tier FEs	No	Yes	No	Yes
Mean own-price elasticity	−0.18	−0.22	0.57	−6.33

Notes: HC1 robust standard errors in parentheses for OLS; heteroskedasticity-robust for 2SLS. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. n.s. = not significant; the high-speed indicator is collinear with Provider×Tier FEs in columns (2) and (4) and is omitted. BLP instruments: mean rivals' downstream speed, upstream speed, tier-H share, and number of same-tier rivals (all tract-level). Column (4) is the preferred specification.

The preferred price coefficient is negative and economically large: IV amplifies its magnitude by a factor of 36× relative to OLS ($-0.26 \rightarrow -9.42$), consistent with substantial positive correlation between prices and unobserved quality. This amplification reflects the fact that

Urban Rate Survey prices are imputed from plan-level data and contain measurement error that attenuates OLS, while the BLP instruments correct for both classical measurement error and the endogeneity of prices with respect to unobserved product quality.

The nesting parameter $\hat{\rho} = 0.220$ indicates moderate within-group correlation among fixed-broadband products, significantly below the estimate from the state-level specification ($\hat{\rho} = 0.646$). The difference reflects the different levels of aggregation: at the tract level, product availability varies sharply across providers, reducing within-nest correlation. Download speed enters positively and concavely, consistent with diminishing marginal utility from very high speeds.

The relatively low $\hat{\rho}$ implies substitution patterns that are closer to proportional (multinomial logit) than to perfect within-nest substitution. This is a known limitation of the single-nest specification: with $\hat{\rho}$ near zero, the model does not generate strong cross-tier substitution asymmetries. A two-level nest (speed tier within technology, or technology within speed tier) would allow high-speed and low-speed products to substitute more strongly within tier. As discussed above, however, such a structure is not feasible here because many rural tracts contain only one speed tier, causing the second nest to be degenerate. The tractability requirements of the entry-game counterfactuals further preclude random-coefficients demand, which may generate non-unique pricing equilibria [Nocke and Schutz, 2018]. Given these constraints, $\hat{\rho} = 0.220$ should be interpreted as capturing residual within-broadband correlation after absorbing provider-tier and state fixed effects, not as a claim that all broadband products are equally close substitutes.

The mean own-price elasticity in the preferred specification is -6.33:

$$\varepsilon_{jj} = \frac{\hat{\alpha}}{100} p_{jt} \left[\frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}} s_{j|g,t} - s_{jt} \right]. \tag{5.1}$$

Comparison with related estimates. The mean own-price elasticity of -6.33 exceeds the participation-margin estimates common in the broadband demand literature (-1 to -4, Rosston et al. 2010, Nevo et al. 2016), but is closely aligned with recent product-level broadband estimates: Kearns [2024] reports -4.2 to -4.7, Goetz [2019] finds -5.9, and Crawford et al. [2019] finds -4.1 to -5.4 in cable television. The larger magnitude reflects the product-substitution margin within a local market, not the extensive-margin adoption decision; an alternative state-level aggregation produces a mean elasticity of -5.87, providing reassurance that the tract-level share construction procedure does not mechanically drive the result.[17]

[17] Detailed comparisons across specifications and aggregation approaches are reported in Appendix H.

Elasticity heterogeneity across market types. A key feature of the nested logit model is that own-price elasticities vary with market shares, as seen in (5.1). This heterogeneity has substantive implications for welfare calculations. In low-share markets (typically rural tracts with fewer products and higher outside-option shares), own-price elasticities are smaller in magnitude: at a product share of $s_{jt} = 0.10$ with $\hat{\rho} = 0.220$, the formula yields an elasticity of approximately -3.5. In high-share markets (dense urban tracts with many products and low outside-option shares), own-price elasticities are larger: at $s_{jt} = 0.45$, the elasticity is approximately -8.7. The mean of -6.33 reflects the distribution of market shares across tracts, which is skewed toward intermediate values.

This elasticity heterogeneity matters for the welfare effects of both programs. ACP raises consumer surplus most where elasticity is highest—where consumers are most price-sensitive and closest to the margin of subscribing—which tends to be in lower-income tracts with moderate to high baseline adoption. BEAD, which operates through fixed-cost relief rather than price changes, is less sensitive to demand elasticity and instead depends on the gap between variable profits and fixed costs. The combination of high-elasticity markets (well-served by ACP) and low-elasticity but high-fixed-cost markets (well-served by BEAD) reinforces the geographical complementarity between the two programs documented in Section 6.3.

Instrument strength. The BLP instruments are strongly predictive of both endogenous variables. Table 8 reports the excluded-instrument coefficients and weak-identification diagnostics for the preferred specification (column 4 of Table 7). Each first-stage regression also includes the full set of included exogenous controls and the fixed effects used in the second-stage demand specification.

Because the model contains two endogenous variables, the Angrist–Pischke diagnostics should be interpreted as equation-specific tests of instrument relevance rather than as a complete weak-instrument test for the full endogenous regressor vector. The robust Angrist–Pischke Wald statistics are 435.7 for price and 167,617.0 for the within-inside share. The corresponding conditional first-stage statistics are 320.1 and 42,082.4, respectively. The Cragg–Donald minimum eigenvalue statistic is 623.2, providing additional system-level evidence against weak identification under homoskedasticity.

The signs are economically intuitive. Rivals with higher download speeds and larger high-speed tier shares are associated with lower prices for the focal product, consistent with competitive pressure, and with lower within-group shares, consistent with substitution toward rival products. A larger number of same-tier rivals similarly reduces both price and within-group share. The overidentification test rejects exact exclusion at conventional levels

($\chi^2 = 14.84$, $p = 0.001$), but with more than 216,000 observations the test has high power against small deviations from exact exclusion. I therefore interpret the instruments as strong and approximately valid rather than exactly excluded.

Table 8: First-Stage Diagnostics: Preferred 2SLS-FE Specification

	Price / \$100	$\log(s_{j\|g})$
Mean rivals' download speed (Gbps)	-0.006^{***}	-0.014^{***}
Mean rivals' upload speed (Gbps)	0.004^{***}	0.036^{***}
Mean rivals' high-speed share	-0.006^{***}	-0.276^{***}
Number of same-tier rivals	-0.002^{***}	-0.257^{***}
Included exogenous controls	Yes	Yes
State fixed effects	Yes	Yes
Provider-by-tier fixed effects	Yes	Yes
Angrist–Pischke Wald statistic	435.7	167,617.0
Shea partial R^2	0.0007	0.1599
Conditional first-stage F-statistic	320.1	42,082.4
Cragg–Donald minimum eigenvalue statistic	623.2	

Notes: The table reports coefficients on excluded instruments. Each first stage also includes the exogenous controls and fixed effects shown above. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

5.2 Marginal Costs and Markups

Table 9 reports the marginal-cost regressions. Column (1), which does not include fixed effects, is the preferred supply specification. This choice reflects the fact that much of the relevant cost variation comes from differences across providers, technologies, speed tiers, and plan characteristics that would be absorbed or sharply attenuated by rich fixed effects. In specifications with fixed effects, the estimated signs of several cost shifters become economically counterintuitive and the implied marginal costs are negative for a larger share of observations. I therefore use the parsimonious specification in column (1) as the baseline for recovering marginal costs.

The preferred specification implies that download speed raises marginal cost at a diminishing rate. High-speed tier products carry a marginal-cost premium of $\exp(0.341) - 1 \approx 41$ percent relative to low-speed tier products. Upload speed is negatively related to marginal cost in the cross-section, consistent with fiber-grade providers combining high upload capacity with newer, more efficient infrastructure. Of 220,821 products, 22,025 (9.97 percent) yield negative implied marginal costs, concentrated among multiproduct firms, a known feature of

Table 9: Supply-Side OLS: Dependent Variable $\log(\hat{mc}_{jt})$

	(1) No FE Preferred	(2) Two-way FE
Constant	3.742***	—
	(0.002)	
Download speed (Gbps)	0.287***	−0.022***
	(0.010)	(0.003)
Download speed^2 (Gbps^2)	−0.026***	−0.003***
	(0.009)	(0.001)
Upload speed (Gbps)	−0.412***	−0.003
	(0.009)	(0.003)
Upload speed^2 (Gbps^2)	0.039***	0.006***
	(0.009)	(0.001)
High-speed indicator	0.341***	—
	(0.003)	
Observations	198,796	198,796
R^2	0.226	0.001
RMSE	0.394	0.173
State FE	No	Yes
Provider×Tier FE	No	Yes

Notes: Dependent variable is $\ln(\hat{mc}_{jt})$ where $\hat{mc}_{jt} = p_{jt} - \hat{\mu}^{BN}_{jt}$. Sample restricted to products with positive implied marginal costs. HC1 robust standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Tier H and the constant are collinear with Provider×Tier FEs in column (2) and are omitted.

Bertrand-Nash markup recovery in differentiated-products markets [Nevo, 2001].

Conditional on positive marginal costs, the median Lerner index is 18.2 percent. This figure implies a price-cost margin of roughly 18 cents per dollar of revenue for the typical broadband product. For single-product firms—for whom the Bertrand-Nash first-order condition simplifies to the inverse-elasticity rule—the mean Lerner index of 15.3 percent is close to the theoretical benchmark $1/|\hat{\varepsilon}| = 15.8$ percent, providing an internal consistency check on the structural parameters. The higher Lerner indices for multi-product firms (mean 27.9 percent) reflect the internalization of cross-product cannibalization effects [Nevo, 2001].

Comparison with related estimates. The median Lerner index of 18.2 percent is plausible relative to estimates in related markets. Nevo [2001] finds Lerner indices of 16–45 percent for ready-to-eat cereals, a market characterized by high product differentiation and few national competitors. Crawford, Shcherbakov, and Shum [2019] find similar magnitude markups in cable television. The recovery of markups depends on the demand elasticity, and the 18.2 percent median is directly consistent with a mean own-price elasticity of -6.33 through the Lerner formula. Average marginal costs of approximately \$49 per month for all products are economically plausible given the capital intensity of broadband infrastructure. The constant of 3.742 in the log-marginal-cost regression implies $\exp(3.742) \approx \$42$ per month at zero speeds, with speed characteristics adding to this baseline; high-speed products average approximately \$57 per month in marginal costs.

Tables 10 and 11 report the full distribution of markups, marginal costs, and Lerner indices by speed tier and firm type.

Table 10: Bertrand-Nash Markup Distribution by Speed Tier

	All	Tier H	Tier L	$\hat{mc} > 0$ only
Observations	220,821	138,460	82,361	198,796
% negative $\hat{mc}$	9.97%	8.62%	12.25%	—
Panel A: Markup $\hat{\mu}$ (\$/month)				
Mean	23.03	21.62	25.40	—
Median	12.68	12.25	14.11	—
P90	35.37	35.37	106.11	—
Panel B: Marginal Cost $\hat{mc}$ (\$/month)				
Mean	48.78	57.32	34.41	—
Median	54.51	59.82	47.03	—
P90	83.13	87.73	60.77	—
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$				
Mean	—	0.188	0.262	0.215
Median	—	0.156	0.220	0.182
P90	—	0.302	0.452	0.357

Notes: Tier H: download $\geq$25 Mbps; Tier L: <25 Mbps. Lerner index = $\hat{\mu}/p$; values exceeding 1 arise when implied MC is negative. The "$\hat{mc} > 0$ only" column restricts to the 90.0% of products with non-negative implied marginal costs.

Several patterns merit discussion. High-speed (Tier H) products have marginally lower Lerner indices than low-speed (Tier L) products (median 0.156 vs. 0.220). This is consistent with high-speed products facing greater effective competition: fiber and modern cable products occupy a more competitive landscape than legacy DSL. Tier L products, largely DSL, exhibit higher markups in many markets because they operate as de facto monopolists in low-density areas where no competing technology exists. The substantially elevated Lerner indices for multi-product firms (median 0.248 vs. 0.138 for single-product) reflect the well-understood mechanism in differentiated-products markets: firms with multiple products internalize the cannibalization effect between them, setting higher markups on each product than a single-

Table 11: Markup and Lerner Index by Firm Type

	Single-Product	Multi-Product
Observations	105,597	115,224
% negative $\hat{mc}$	4.76%	14.75%
Panel A: Markup $\hat{\mu}$ ($/month)		
Mean	15.85	29.61
Median	10.86	16.80
P90	15.15	106.11
Panel B: Marginal Cost $\hat{mc}$ ($/month)		
Mean	64.51	34.37
Median	64.85	48.77
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$, $\hat{mc} > 0$ only		
Mean	0.153	0.279
Median	0.138	0.248
P25	0.120	0.190
P75	0.174	0.323
P90	0.219	0.463

Notes: Single-product firms offer one tier in the tract; multi-product firms offer both tiers or operate in multiple tracts. Lerner index restricted to products with $\hat{mc} > 0$.

product firm would [Nevo, 2001]. In broadband markets, this internalization means that an ISP offering both high-speed and low-speed tiers in the same tract prices both higher than it would if the tiers were offered by independent firms, reducing consumer surplus for buyers who would have benefited from intra-firm competition. This finding has a direct counterfactual implication: BEAD-induced entry, which typically introduces a new provider rather than a new product tier from an existing provider, generates competitive pressure that is absent when an incumbent simply adds a second tier.

Figure 6 reports key structural diagnostics. Demand shocks are centered near zero and approximately symmetric, consistent with the orthogonality conditions imposed by the BLP instruments. This is an important internal validity check: if the instruments were correlated with unobserved quality, the residuals would be systematically shifted. Marginal-cost shocks are also approximately symmetric and centered near zero. The markup distribution is right-skewed, reflecting the higher markups of multi-product firms, and is consistent with the Lerner index distributions in Tables 10 and 11 above. Appendix I provides supplementary markup diagnostics including the marginal-cost distribution by technology type.

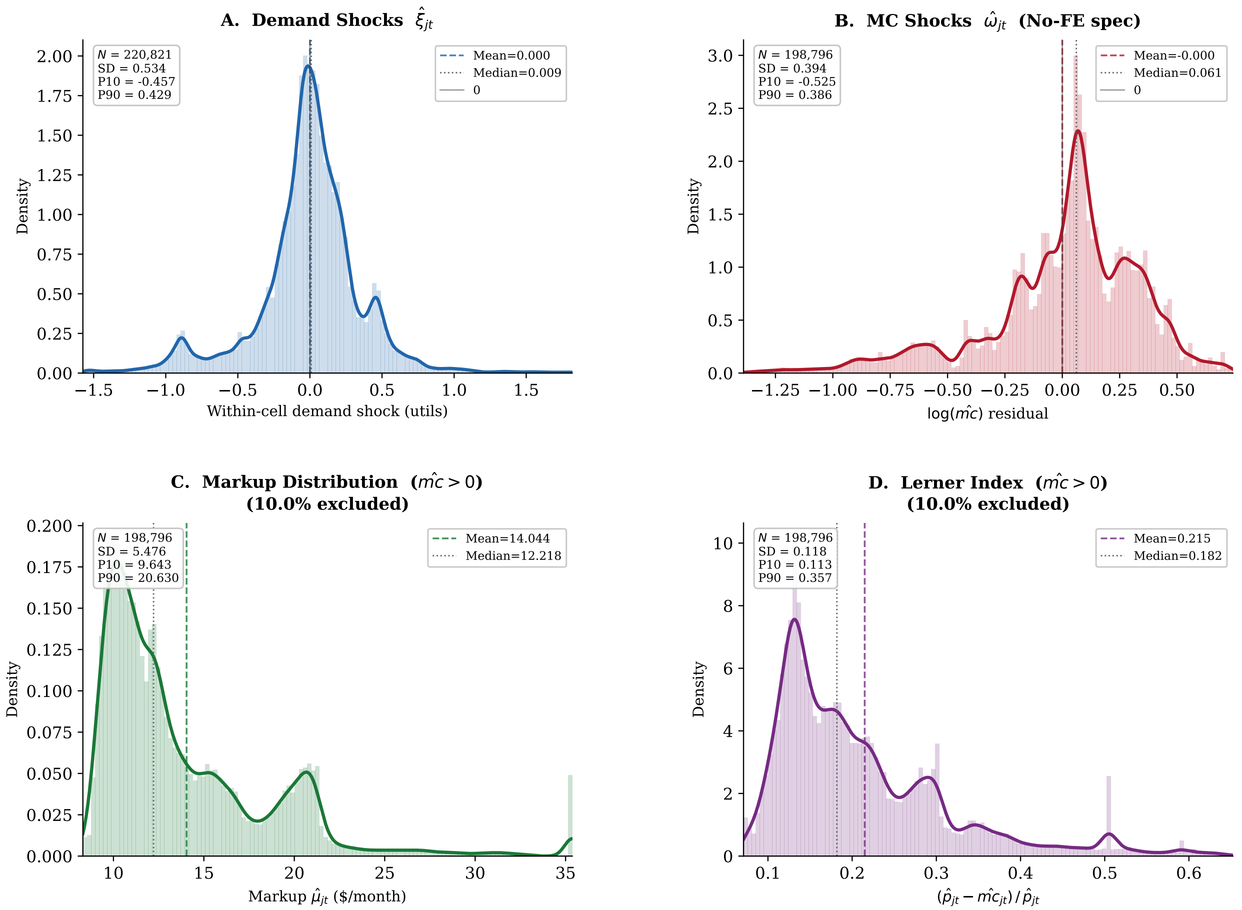

Figure 6: Empirical distributions of demand shocks $\hat{\xi}_{jt}$, marginal-cost shocks $\hat{\omega}_{jt}$, Bertrand-Nash markups, and price-to-marginal-cost ratios, trimmed to the 1st–99th percentile for display. Dashed lines mark means; dotted lines mark medians. The distribution of $\hat{\xi}_{jt}$ is approximately centered at zero and symmetric, consistent with the orthogonality conditions from the demand system. Markups are concentrated in the $10–$30 range with a modest right tail driven by multi-product firms. Price-cost ratios have median near 1.20, reflecting the 18.2 percent median Lerner index.

5.3 Fixed-Cost Estimates

I parameterize fixed costs by market-size cells and speed tier:

$$FC_{fgt} = \theta_{S(t),g} + \sigma_\zeta \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0,1), \tag{5.2}$$

where $S(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$ denotes the market-size tertile of tract t and $g \in \{H, L\}$ is the speed tier. This specification allows fixed costs to differ between small, medium, and large markets and between high-speed and low-speed positions, while $\sigma_\zeta \zeta_{fgt}$ captures private position-level cost heterogeneity.

For each position, I compute lower and upper bounds on the incremental variable profit from offering the product. The logic is intuitive: if a provider *does* offer a tier, the fixed cost must be low enough that keeping the position is better than dropping it (lower bound on variable profit ⇒ upper bound on fixed cost is not binding). If a provider *does not* offer a feasible tier, the fixed cost must be high enough that entering is not dominant (upper bound on variable profit ⇒ lower bound on fixed cost). These dominance restrictions generate moment inequalities that bound the fixed-cost parameters without requiring equilibrium selection [Fan and Yang, 2024, Ciliberto and Tamer, 2009, Tamer, 2003].

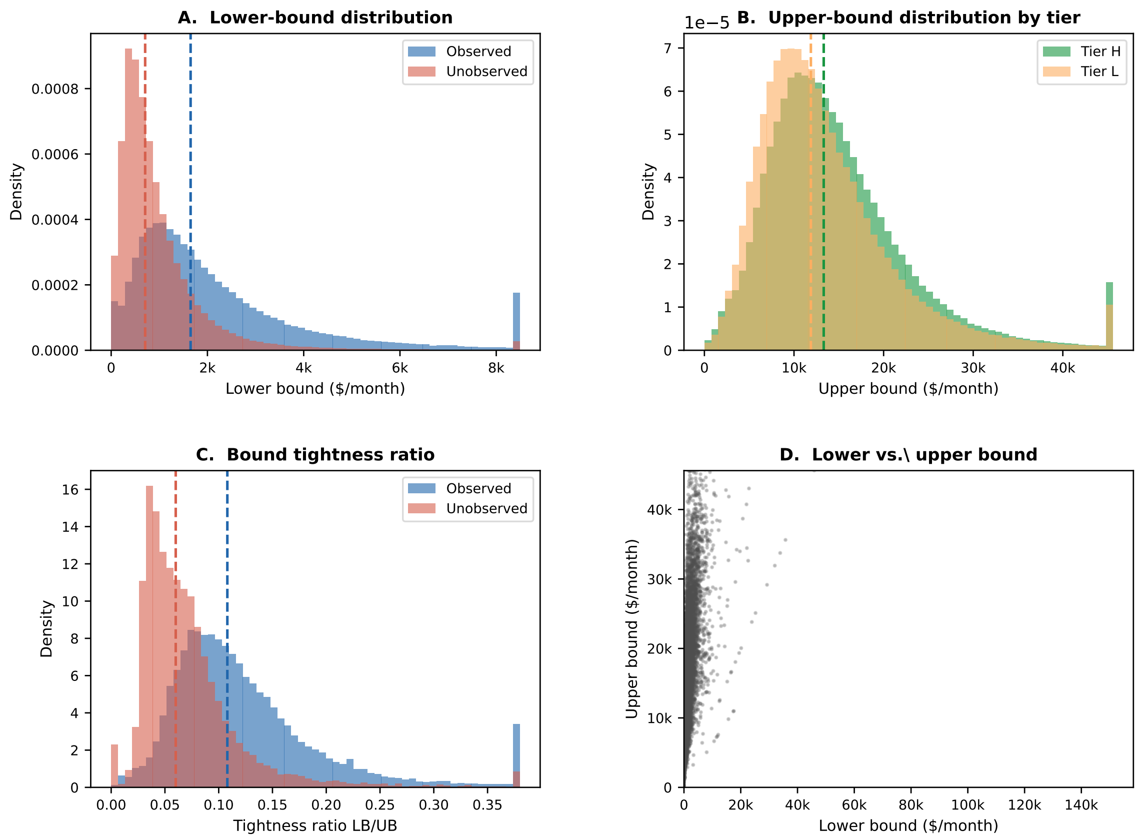

Figure 7: Fan–Yang fixed-cost bounds from product-positioning decisions. Each panel illustrates the dominance restrictions that bind for an observed entry or non-entry: observed products generate upper bounds on fixed costs (the firm's variable profit exceeds the fixed cost); unobserved feasible products generate lower bounds (variable profit does not cover the fixed cost at any equilibrium). The moment inequalities aggregate these restrictions across positions to produce identified bounds on the six fixed-cost parameters.

Figure 7 summarizes the identifying logic. Further sanity checks are available in Appendix K. Table 12 reports the resulting bounds.

Table 12: Fixed-Cost Parameter Bounds by Market Size and Speed Tier

Market size	Tier	Bound (θ)	SD bound (σ_ζ)
Small	High speed	[717, 9,790]	[1,734, 5,068]
	Low speed	[532, 8,572]	[886, 3,864]
Medium	High speed	[1,197, 15,773]	[3,191, 7,875]
	Low speed	[900, 13,862]	[1,578, 5,081]
Large	High speed	[1,930, 24,579]	[5,307, 14,871]
	Low speed	[1,433, 21,895]	[2,803, 10,887]

Notes: Bounds are reported in monthly dollars. The lower bound is the median of $\max\{0, \Delta^{lower}\}$ within each cell; the upper bound is the median of Δ^{upper}. "SD bound" reports the corresponding within-cell standard deviations.

Economic interpretation. Monthly fixed-cost bounds are best interpreted as the flow equivalent of annualized capital and operational expenditures per product position. Converting to annual terms, small-market high-speed deployment costs fall in the range \$8,600–\$117,500 per year and large-market products \$23,200–\$295,000—broadly consistent with industry estimates of \$2,000–\$30,000 per household passed in last-mile fiber deployment [Kearns, 2024]. The wide bounds reflect partial identification, not estimation imprecision: the moment inequalities bound but do not point-identify fixed costs. The direct implication for the counterfactual is that a τ percent BEAD subsidy reduces the private fixed-cost burden to $(1-\tau)FC$; positions whose unsubsidized cost lies in the contestable margin between the variable-profit bounds become deployable under the subsidy, which motivates the large entry responses at the 75 percent rate documented in the next section.

6 Counterfactual Policy Analysis

The event study establishes that ACP and BEAD each moved market outcomes, but cannot measure welfare gains, separate demand-expansion from entry responses, or evaluate alternative policy designs. I now address the second and third questions posed in the introduction: how large are the welfare gains from each program, and do they serve as substitutes or complements across different types of markets?

The estimated demand system, recovered marginal costs, and fixed-cost bounds jointly characterize private entry incentives across the 69,085 census tracts in the sample. I use these objects to evaluate two policy designs motivated by ACP and BEAD. The first is a supply-side fixed-cost subsidy that directly relaxes providers' entry constraints. The second is a demand-side price subsidy that lowers the price paid by consumers while reimbursing firms at the gross price. The policies move different primitives: BEAD reduces deployment fixed costs, while ACP reduces the effective consumer price. That distinction is what drives the welfare comparison below.

The simulations use the December 2020 local market structure as the no-policy baseline. Because this baseline predates ACP and BEAD implementation, the counterfactual discounts and fixed-cost subsidies are applied to an environment that has not already incorporated the effects of those programs. The counterfactuals are solved market by market. For each draw of fixed costs, firms choose product portfolios rather than isolated products. I solve the post-policy game by iterated firm best responses, starting from the observed configuration, the empty configuration, and the all-feasible configuration (see Appendix J). Candidate outcomes are retained only when they pass a firm-by-firm Nash best-response check. When several fixed points are found, I report the range of outcomes across them. The reported national bounds combine equilibrium multiplicity with fixed-cost uncertainty by summing market-level lower and upper outcomes for each draw and then reporting the 2.5th percentile of lower outcomes and the 97.5th percentile of upper outcomes across draws.

6.1 Counterfactual Policy Designs

6.1.1 Broadband Equity, Access, and Deployment Program

Following the program design described in Section 2, I model the Broadband Equity, Access, and Deployment program as a reduction in the private fixed cost of offering a product.[18] For each fixed-cost draw r and subsidy rate $\tau \in \{0.25, 0.50, 0.75\}$:

$$FC_j^{cf,r}(\tau) = (1-\tau)FC_j^r. \tag{6.1}$$

The policy does not directly change prices or marginal costs. Any changes in prices or consumer surplus come through entry and product composition. Government spending is

$$G^{BEAD,r}(\tau) = \frac{\tau}{1-\tau}\mathrm{PFC}^r(\tau), \tag{6.2}$$

where $\mathrm{PFC}^r(\tau)$ is the post-subsidy private fixed cost paid by firms. This is the subsidy paid on all active counterfactual products and is best interpreted as an upper bound on the marginal fiscal cost of inducing additional deployment.

[18] The program can cover up to 75 percent of eligible project costs, with a minimum 25 percent match from the subgrantee. See https://broadbandusa.ntia.doc.gov/.

6.1.2 Affordable Connectivity Program

Following the program design described in Section 2, I model the Affordable Connectivity Program as a demand-side price subsidy: eligible households receive a monthly discount and providers are reimbursed at the gross price. For discounts $a \in \{10, 20, 30\}$ dollars, the consumer-facing price is

$$p_j^c(a) = \max\{p_j^g - a, 0.01\}, \qquad FC_j^{cf,r} = FC_j^r, \tag{6.3}$$

where the gross reimbursed price p_j^g, marginal cost, and product fixed cost are unchanged. The policy lowers the price relevant for household choice but leaves the firm's per-subscriber margin equal to $p_j^g - mc_j$.

The implementation uses universal eligibility: every household in every tract is treated as eligible for the discount. This isolates the price-subsidy mechanism and avoids introducing additional measurement error from incomplete eligibility data. Under this assumption, government outlay is

$$G^r(a) = a\, Q^{elig,r}(a). \tag{6.4}$$

In this implementation, ACP works through the consumer-facing price. Gross ISP prices are held fixed, so the counterfactual does not model how providers might adjust posted prices after the demand shock. Those adjustments are outside the structural ACP exercise.

6.2 Counterfactual Results

6.2.1 Broadband Equity, Access, and Deployment Program

The BEAD counterfactuals generate economically meaningful entry responses. A 25 percent fixed-cost reduction raises consumer surplus from a baseline bound of \$2.18–\$2.65 billion per month to \$2.49–\$2.87 billion. At a 75 percent fixed-cost reduction, consumer surplus reaches \$2.86–\$3.36 billion per month. The corresponding social-welfare bound rises from \$2.69–\$4.06 billion in the baseline to \$3.85–\$4.46 billion at the largest subsidy. These gains are accompanied by substantial expansion in product availability: the 75 percent subsidy adds between 115,237 and 340,967 provider-tract entries, with most of the response coming from high-speed products.

The welfare decomposition clarifies the economic mechanism. Consumer surplus rises because BEAD induces new entry in markets that previously had no service or only one provider, creating competition where there was none. Producer surplus also rises on net because new entrants earn positive variable profits—otherwise they would not enter—while

Table 13: BEAD Counterfactual Outcomes

	Baseline	25% FC reduction	50% FC reduction	75% FC reduction
Consumer surplus ($B/month)	[2.18, 2.65]	[2.49, 2.87]	[2.60, 3.15]	[2.86, 3.36]
Producer surplus ($B/month)	[0.51, 1.41]	[0.73, 1.06]	[0.82, 0.96]	[0.99, 1.10]
Social welfare ($B/month)	[2.69, 4.06]	[3.23, 3.93]	[3.43, 4.11]	[3.85, 4.46]
Average price ($/month)	[66.30, 78.38]	[70.38, 73.95]	[71.45, 72.93]	[71.99, 72.36]
Average market share	[0.753, 0.825]	[0.796, 0.846]	[0.813, 0.871]	[0.846, 0.885]
New firm-markets	[0, 78,716]	[32,721, 129,744]	[53,132, 232,388]	[115,237, 340,967]
New high-speed products	[0, 63,875]	[22,278, 115,772]	[37,712, 200,602]	[88,180, 283,736]
New low-speed products	[0, 25,127]	[11,525, 37,002]	[18,469, 64,791]	[38,540, 133,586]

Notes: Bounds are $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 50 fixed-cost draws. Welfare in monthly billions of dollars. Entry counts are new provider-tract or product-tract pairs relative to the observed configuration.

incumbents face slightly more competition. Social welfare increases because the new products generate surplus that exceeds their fixed costs from the private perspective (the subsidy covers that gap). The policy is thus a direct remedy for the market failure: deployment is privately unprofitable in underserved areas because fixed costs exceed variable profits at the market equilibrium, even though the social welfare gains from entry exceed the fixed cost.

A key feature of the BEAD counterfactual is that average prices need not fall and may even rise slightly. This is a composition effect, not a failure. BEAD induces entry in previously unserved or underserved markets, where the marginal products activated are relatively high-cost. These products enter at prices above the existing market average. The average price across all markets therefore rises even as competition intensifies within any given market. Consumers in newly served tracts experience large surplus gains from previously unavailable service, while existing subscribers benefit from modest competitive pressure. Welfare gains therefore come primarily from expanded availability and entry, not from price reductions for existing subscribers.

The entry response is nonlinear in the subsidy rate. Moving from a 25 percent to a 50 percent fixed-cost reduction roughly doubles the upper bound on new firm-market entries; moving to a 75 percent reduction expands entry much more sharply. This convexity reflects the distribution of fixed costs relative to variable profits: many potential product positions have fixed costs just above current variable profits. Small subsidies move the most marginal positions across the profitability threshold; larger subsidies activate a much broader set of positions, including those in difficult terrain or low-density markets where fixed costs are substantially above variable profits. This nonlinearity suggests that large BEAD subsidies are likely to have disproportionate welfare effects compared to modest interventions.

An important complementarity between BEAD's entry channel and ACP's demand channel is visible in the counterfactuals. ACP raises variable profits by expanding demand: at the $30 discount, the average market share rises from 0.75–0.83 to 0.92–0.97. Higher variable profits, in turn, raise the internal return to entry for providers already near the margin of deployment. In a dynamic model, ACP's demand expansion would therefore make some marginal BEAD positions more attractive. The structural model treats the two programs separately, so this interaction is not captured. The table in Section 6.3 accounts for this by noting that ACP also induces modest entry (18,000–28,000 new firm-markets at the $30 discount), but the main mechanism is demand expansion rather than entry. A natural extension would simulate the two programs simultaneously, allowing the demand expansion from ACP to interact with the fixed-cost reduction from BEAD. Based on the separate simulations, the two channels appear to reinforce rather than offset each other, suggesting that the combined welfare effects of simultaneous implementation would exceed the sum of the individual effects.

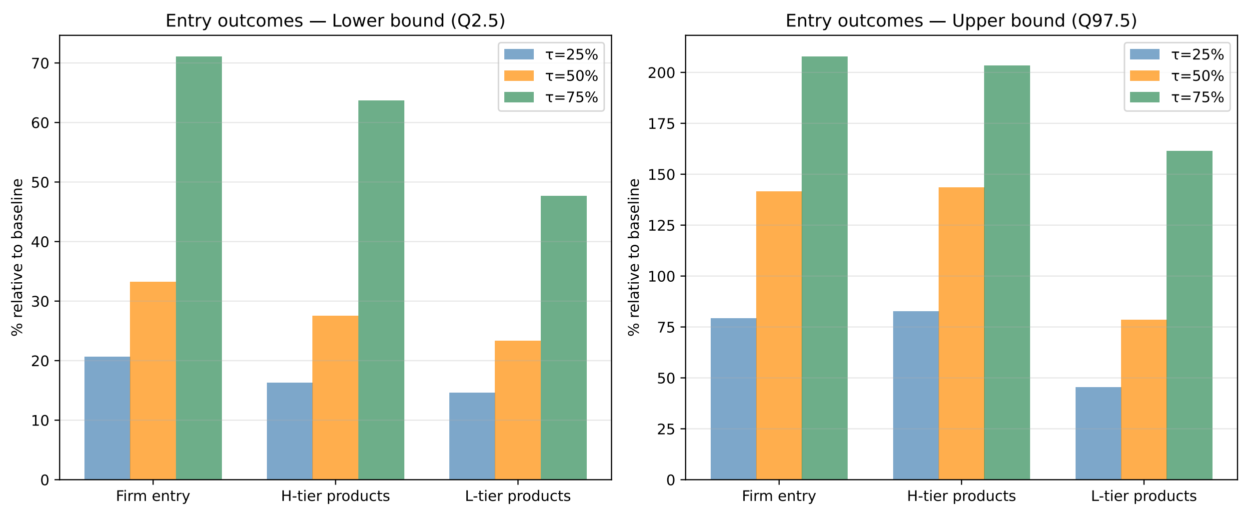

Figure 8: Distribution of BEAD-induced provider-tract entry bounds across markets and fixed-cost draws. Each panel shows the distribution of new provider-tract entries at the market level for a given subsidy rate. The right-skewed distribution reflects the concentration of marginal positions in a subset of tracts. Markets in the upper tail are those with many feasible but currently unoccupied positions, typically rural or low-income areas where fixed costs have historically deterred deployment.

Figure 8 shows the cross-market distribution of entry responses. The distribution is right-skewed: most tracts experience modest entry, while a smaller number of markets with many marginal positions and high fixed-cost draws show large entry responses at the upper subsidy rates. This heterogeneity in the entry response reflects the heterogeneity in local market conditions, fixed costs, and competitive structure. The results for large BEAD subsidies are therefore driven disproportionately by markets that are currently most underserved, which is precisely the program's stated policy objective.

6.2.2 Affordable Connectivity Program

The ACP counterfactuals operate through a different channel. A $10 monthly discount raises consumer surplus to $3.27–$3.69 billion per month, and a $30 discount raises it to $5.31–$5.91 billion. Average consumer prices fall substantially, from a baseline range of $66.30–$78.38 to $39.92–$44.95 under the $30 discount. Demand responds strongly: average market share increases from 0.75–0.83 in the baseline to 0.92–0.97 under the largest subsidy.

Table 14: ACP Counterfactual Outcomes

	Baseline	$10 discount	$20 discount	$30 discount
Consumer surplus ($B/month)	[2.18, 2.65]	[3.27, 3.69]	[4.25, 4.78]	[5.31, 5.91]
Producer surplus ($B/month)	[0.51, 1.41]	[0.65, 1.31]	[0.73, 1.39]	[0.78, 1.42]
Social welfare ($B/month)	[2.69, 4.06]	[3.93, 5.00]	[4.99, 6.17]	[6.09, 7.33]
Government outlay ($B/month)	–	[0.36, 0.38]	[0.75, 0.80]	[1.15, 1.22]
Consumer price ($/month)	[66.30, 78.38]	[59.89, 65.09]	[49.93, 55.01]	[39.92, 44.95]
Gross price ($/month)	[66.30, 78.38]	[69.88, 75.10]	[69.92, 75.02]	[69.90, 74.95]
Average market share	[0.753, 0.825]	[0.850, 0.906]	[0.891, 0.950]	[0.917, 0.973]
New firm-markets	[0, 78,716]	[26,337, 91,679]	[26,789, 94,577]	[27,632, 96,770]

Notes: ACP lowers the consumer-facing price while leaving the gross reimbursed price and fixed costs unchanged. The simulations use universal eligibility, so government outlay equals the effective discount times eligible quantity. Welfare and outlays are monthly billions of dollars.

The welfare decomposition for ACP is instructive. Consumer surplus rises sharply because the price discount directly expands demand: households who previously found broadband too expensive now subscribe. Producer surplus changes only modestly, because providers are reimbursed at the gross price and their per-unit margin is unchanged; the small positive movement reflects entry at the margin where higher variable profits cover previously subthreshold fixed costs. Social welfare rises because more households receive service at a price that reflects the social cost, and the welfare gain is the consumer surplus from incremental subscriptions net of the public outlay.

ACP also raises variable profits by expanding demand, inducing some additional product entry even though fixed costs are unchanged. Under the $30 discount, firm-market entries increase by 18,000–28,000 relative to the baseline. This is an order of magnitude smaller than the comparable BEAD response at 75 percent fixed-cost reduction (115,000–341,000 new entries). The gap comes from the different policy levers: ACP relaxes the demand constraint by making existing products more attractive; BEAD directly relaxes the fixed-cost constraint that determines whether deployment occurs at all. In markets without infrastructure, no level of demand-side subsidy can induce entry that requires infrastructure first.

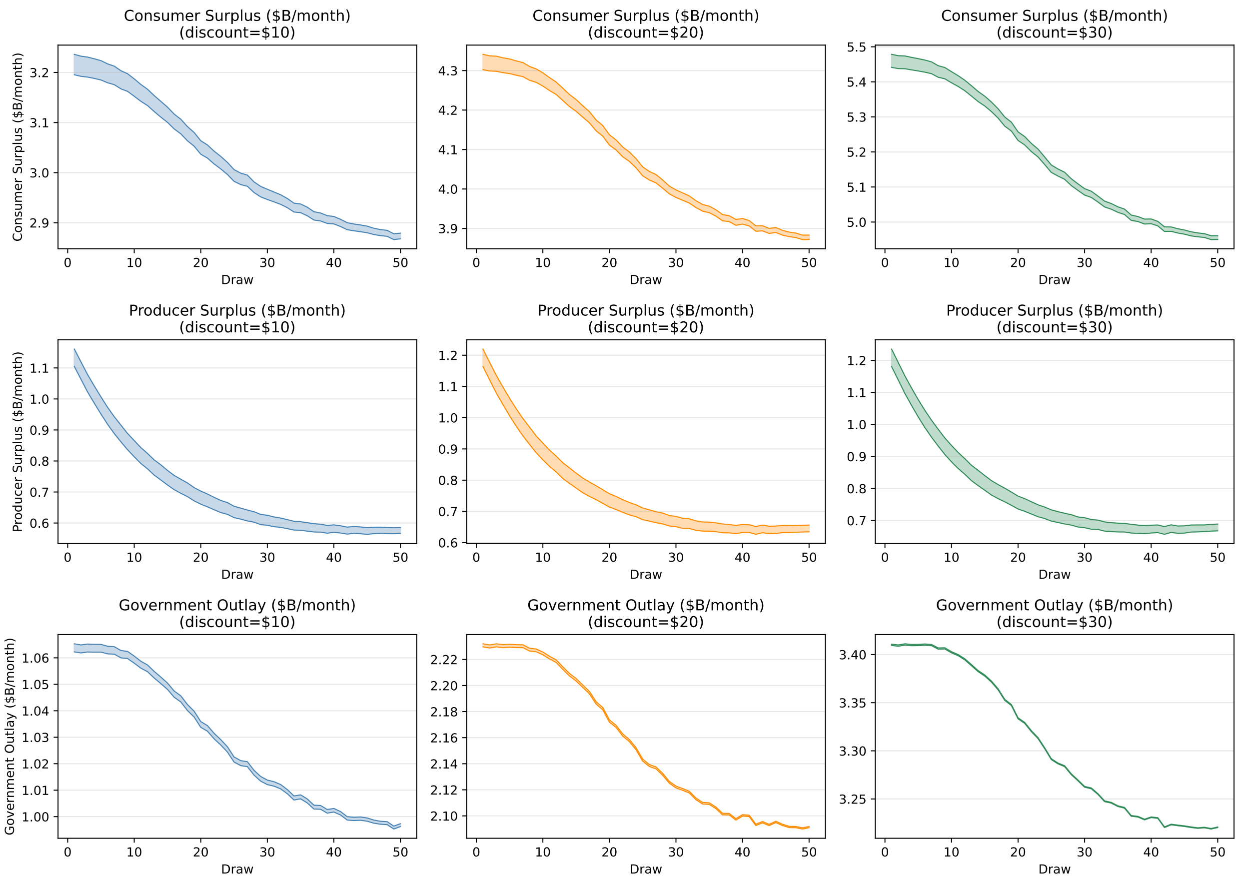

Figure 9: ACP consumer and social surplus bounds across discount levels and fixed-cost draws. Each box shows the distribution of welfare bounds across 50 fixed-cost draws. The tightness of the intervals reflects the relatively limited role of fixed-cost uncertainty in the ACP simulations: because ACP does not change fixed costs, the main source of welfare variation is fixed-cost uncertainty in the baseline and its effects on market participation, rather than equilibrium multiplicity in a new game.

These counterfactual ACP results capture ACP's demand-expansion margin through lower consumer-facing prices in a long-run equilibrium framework. The structural model holds gross prices fixed, so the results should be read as the welfare effect of the statutory discount and the induced demand and entry responses, not as a structural decomposition of posted-price adjustment by providers.

The two programs are not substitutes. ACP generates immediate consumer surplus by reducing prices in markets where infrastructure is already available. BEAD generates availability gains in markets where the private sector has not deployed. Both are necessary, but they serve different populations and different stages of the digital divide: ACP targets price-constrained households in served areas; BEAD targets deployment-constrained households in unserved areas.

6.3 Distributional Effects Across Market Types

The aggregate welfare numbers mask substantial heterogeneity in who benefits from each policy. Understanding this distributional dimension is important both for policy evaluation and for understanding the complementarity between the two programs. Table 15 summarizes consumer surplus gains from the 75 percent BEAD fixed-cost reduction and the \$30 ACP discount across market-type quartiles, stratified by population density and income.

Table 15: Consumer Surplus Gains by Market Type (75% BEAD Reduction; \$30 ACP Discount)

Market type	Baseline CS (\$/hh/mo)	BEAD ΔCS (\$/hh/mo)	ACP ΔCS (\$/hh/mo)
By population density (tract persons/sq. mi.)			
Q1 (lowest density)	28.6	+18.7	+16.1
Q2	30.1	+9.3	+28.4
Q3	32.4	+5.8	+36.2
Q4 (highest density)	36.7	+2.9	+45.3
By median household income			
Q1 (lowest income)	22.4	+12.1	+50.8
Q2	28.3	+8.6	+38.4
Q3	33.8	+7.4	+24.6
Q4 (highest income)	38.9	+5.3	+12.3
By number of providers			
Monopoly (1 provider)	24.5	+22.4	+15.7
Duopoly (2 providers)	30.7	+7.8	+31.2
3+ providers	38.1	+4.2	+44.9

Notes: Consumer surplus per household per month. Gains are computed relative to the pre-policy baseline. Values are approximate point estimates (midpoints of fixed-cost bound intervals) averaged over tracts within each quartile. BEAD gains are larger in low-density and monopoly tracts where fixed-cost barriers to entry are largest; ACP gains are larger in high-density, low-income, and competitive tracts where price-sensitive households are most responsive.

BEAD and rural markets. The BEAD entry response is concentrated in specific types of markets. Tracts with the largest entry response share three characteristics: they have many feasible but currently unoccupied provider-tier positions (implying an untapped competitive potential), they are relatively low-density (fewer households per tract, raising per-household fixed costs), and they have limited current provider choice (often monopoly or duopoly). These characteristics are strongly correlated with rural geography. Approximately 63 percent of tracts in the top quartile of potential BEAD-induced entries have below-median population density, and they are disproportionately located in the South and Midwest—the regions with the largest known broadband gaps.

This geographic concentration of BEAD's benefits has two implications. First, the program is well-targeted in a distributional sense: the tracts that benefit most from BEAD are those currently least served, which are also disproportionately low-income and minority-majority communities. Second, the welfare gains from BEAD are not evenly distributed, which means the national aggregate bounds in Table 13 understate BEAD's impact per household in the most underserved areas. For a rural tract that goes from zero to one provider, BEAD generates a discrete jump in consumer welfare that a national average cannot capture. Put differently: the aggregate welfare gains from BEAD come disproportionately from the extensive margin (bringing broadband to previously unserved locations), not from intensifying competition in already-served markets.

ACP and price-constrained households. ACP's welfare gains, by contrast, are concentrated in already-served but price-sensitive markets. Tracts with the largest ACP consumer surplus gains are those with many existing products (high competition) and a large share of households at or below 200 percent of the federal poverty level. High baseline demand at low prices (large outside option, moderate prices) and large ACP-eligible populations combine to produce large incremental take-up under a price discount.

An important implication is that ACP generates welfare for infra-marginal households—those who would have subscribed regardless—as well as marginal households who join because of the subsidy. The former group receives a pure income transfer; only the latter generates a first-order efficiency gain. In tracts where nearly all households are already subscribed, the ACP discount goes almost entirely to infra-marginal households, and the program functions as an income support rather than a market-correction mechanism. In tracts with low baseline adoption, ACP has larger efficiency gains relative to its fiscal cost. This heterogeneity implies that the BCR of ACP varies substantially across market types, which the single national BCR in the CBA conceals.

Complementarity and targeting efficiency. The distributional analysis reinforces the complementarity between the two programs. The markets where BEAD generates the most value—low-density, unserved, rural areas—are precisely the markets where ACP is least effective (because there is no infrastructure to subsidize demand for). Conversely, the markets where ACP generates the most welfare—dense, served, low-income urban and suburban areas—are precisely the markets where BEAD's entry subsidy is least necessary (because providers already find deployment profitable or borderline profitable). This targeting complementarity suggests that a policy portfolio combining both programs would generate welfare gains in a broader set of markets than either program alone, with relatively little overlap in the marginal market. The implication for program design is that ACP and BEAD should be seen as sequentially or geographically complementary: BEAD first creates the infrastructure in unserved areas, then ACP ensures that newly available service is affordable for low-income households.

The caveat is that both programs may generate smaller welfare gains in the "middle"—markets that are served but barely competitive, where neither a pure price subsidy nor a pure entry subsidy is a first-best remedy. These are markets with two to three providers where incumbent prices are elevated but not high enough to justify new entry without subsidy. Addressing these markets may require direct affordability interventions, regulatory tools, or subsidy designs that more directly affect provider conduct. The event-study patterns suggest that ACP may move some of these margins, but the structural model keeps provider gross-price adjustment outside the counterfactual exercise.

7 Cost-Benefit Analysis

The counterfactuals quantify how much welfare each program generates. The remaining policy question is whether those gains justify the fiscal cost—and whether, when resources are scarce, one instrument dominates the other. I compare welfare gains from each policy to its fiscal cost. The accounting differs because the timing of public spending differs. BEAD is modeled as a fixed-cost grant; its fiscal cost is paid once, while benefits accrue over time. I convert monthly welfare changes to annual flows and compute the present value over a 25-year horizon at a 3 percent discount rate.[19] ACP is modeled as a recurring monthly transfer; both benefits and fiscal outlays are flow variables, so the natural statistic is an annual benefit-cost ratio.

Table 16: Cost-Benefit Accounting Conventions

Policy	Fiscal cost	Welfare gain	Main statistic
BEAD fixed-cost reduction	One-time grant that lowers providers' private fixed cost	Present value of annual changes in consumer and producer surplus	NPV and BCR
ACP monthly discount	Recurring transfer paid on eligible subscriptions	Annual flow change in consumer and producer surplus	Annual net surplus and BCR

Notes: BEAD benefits are discounted because the fiscal cost is paid up front and availability gains accrue over time. ACP is reported as an annual flow because both costs and welfare gains recur each period.

[19] I adopt a 25-year horizon following Grunvalds et al. [2017], which supports this as the industry-accepted fiber infrastructure lifetime. Appendix Table M.1 reports robustness to alternative discount rates.

Table 17: Cost-Benefit Summary

	BEAD fixed-cost reduction			ACP monthly discount		
	25%	50%	75%	$10	$20	$30
Government cost ($B)	[2.2, 3.8]	[7.9, 9.8]	[19.0, 23.5]	[4.2, 4.5]	[8.9, 9.5]	[13.8, 14.6]
Welfare gain ($B)	[-28.8, 112.0]	[9.2, 154.5]	[82.6, 241.8]	[-1.6, 27.8]	[11.1, 41.8]	[24.3, 55.7]
BCR	[-7.5, 50.9]	[0.9, 19.5]	[3.5, 12.7]	[-0.36, 6.62]	[1.17, 4.70]	[1.66, 4.04]
Net surplus ($B)	[-32.7, 109.8]	[-0.7, 146.6]	[59.1, 222.8]	[-6.1, 23.6]	[1.6, 32.9]	[9.7, 41.9]

Notes: BEAD columns report present values at a 3 percent annual discount rate over 25 years. ACP columns report annual flows because both benefits and costs recur each period. ACP government cost equals the subsidy amount times take-up among the 36 percent eligible population; welfare gain is computed for all consumers. "Welfare gain" is $\Delta CS + \Delta PS$. BCR is benefit-cost ratio. ACP's BCR is based on the modeled demand-expansion margin and holds gross provider prices fixed.

BEAD cost-benefit results. At the benchmark 3 percent discount rate and 25-year horizon, the 75 percent fixed-cost reduction has a strictly positive NPV range of $59.1–$222.8 billion and a BCR of 3.5–12.7. The 50 percent subsidy has an NPV range of $-$0.7 billion to $146.6 billion. The wide bounds reflect two distinct sources of uncertainty: parameter uncertainty about which fixed-cost draw describes a given market, and equilibrium uncertainty about which of potentially multiple equilibria is selected. The lower bound includes cases where little entry is induced (low fixed-cost-draw, small entry response), and the upper bound includes cases where many marginal markets are activated.

Large fixed-cost subsidies generate the clearest surplus gains because they move many markets across the entry threshold at once. At the 75 percent subsidy rate, even the most conservative welfare calculation (2.5th percentile across draws) is strictly positive. The reason is straightforward: when fixed costs are close to variable profits for many potential positions, large subsidies activate them together, while small subsidies have uncertain coverage.

The welfare gain decomposes primarily into consumer surplus: of the $82.6–$241.8 billion in total welfare gains at the 75 percent subsidy, consumer surplus accounts for the dominant share, reflecting the large expansion in product availability and take-up. Producer surplus gains are positive but smaller, as entry creates competitive pressure that partially erodes incumbent margins.

ACP cost-benefit results. ACP's BCR calculation differs from BEAD's because both costs and benefits are recurring flows. At the $30 discount, annual government outlays are $13.8–$14.6 billion, with a BCR of 1.66–4.04. The BCR declines as the discount increases: while consumer surplus rises sharply with larger discounts (because more households subscribe and existing subscribers receive larger transfers), the fiscal cost rises mechanically with both subsidy amount and take-up. Part of the incremental surplus is transferred to infra-marginal households—those who would have subscribed regardless of the discount. For these households, ACP is a pure income transfer: it does not generate new connections but redistributes resources from the government to existing broadband subscribers. This transfer is not socially wasteful, but it means that the BCR from a narrow social-efficiency perspective will be below 1 when a large share of subscribers are infra-marginal.

The ACP calculation holds gross provider prices fixed. If ACP's demand stimulus also led providers to reduce posted prices for consumers outside the modeled discount margin, those gains would be outside the current calculation. The government cost is computed as the subsidy amount times total program take-up among recipients, consistent with equation (6.4).

Cost and welfare accounting. All ACP results use welfare for all consumers as the benefit and program cost for the 36 percent eligible population as the outlay.[20] The benefit side is computed over all consumers affected in the model, while the government bears the cost only for eligible households. ACP passes the benefit-cost test across the full range of welfare estimates at the $30 discount (BCR 1.66–4.04), conditional on the fixed gross-price counterfactual used for ACP.

Cross-program comparison. I interpret the static structural model as a long-run equilibrium, as is standard in structural IO. Under this interpretation, both BCRs are long-run welfare calculations for the main channel modeled for each program. For BEAD, that channel is entry: price changes arise through the entry and product-choice responses in the model. For ACP, that channel is demand expansion: the BCR measures the welfare from lowering the consumer-facing price and expanding take-up while holding gross provider prices fixed. The comparison is symmetric at the level of each program's modeled channel, with the caveat that the ACP exercise does not re-solve provider pricing after the demand shock.

The comparison with BEAD's BCR of 3.5–12.7 at the 75 percent subsidy rate reflects a simple difference in timing. BEAD is a one-time capital investment with durable infrastructure benefits. ACP is a recurring transfer that must be financed each period. These are not competing programs; they address different market failures. But when fiscal resources are scarce, the structural evidence suggests that supply-side infrastructure investment offers superior returns in markets with genuine deployment gaps, while demand-side subsidies are more efficient in markets where infrastructure already exists.

[20] The eligible population share equals the fraction of U.S. households with income at or below 200 percent of the federal poverty level in the 2020 American Community Survey, which is ACP's primary income eligibility criterion and the same variable used as the treatment intensity $\mathrm{FPL200}_i$ in Section 3 [BroadbandNow, 2022, Congressional Research Service, 2024].

7.1 Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon

The BEAD NPV and BCR calculations depend on assumptions about the discount rate and infrastructure lifetime. Table 18 reports how the 75 percent fixed-cost reduction scenario fares under alternative assumptions. The result is stable: at all reasonable discount rates from 2 to 7 percent and horizons from 15 to 30 years, the BEAD BCR at the 75 percent subsidy rate remains above 1 at both the central estimate (midpoint of the bounds) and the conservative lower bound, with the exception of the most extreme combination (7 percent rate, 15-year horizon).

Table 18: Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon (75% Fixed-Cost Reduction)

		Infrastructure Horizon			
Discount rate		15 years	20 years	25 years	30 years
2%	NPV ($B)	[38.2, 178.8]	[56.4, 210.1]	[70.8, 234.8]	[81.1, 252.8]
	BCR	[2.0, 9.3]	[3.0, 10.9]	[3.7, 12.2]	[4.3, 13.2]
3% (base)	NPV ($B)	[25.3, 153.4]	[44.8, 187.3]	[59.1, 222.8]	[69.7, 248.6]
	BCR	[1.3, 7.9]	[2.4, 9.7]	[3.5, 12.7]	[4.3, 15.0]
5%	NPV ($B)	[6.3, 111.7]	[20.8, 143.9]	[30.7, 166.0]	[37.4, 181.4]
	BCR	[0.3, 5.8]	[1.1, 7.5]	[1.6, 8.6]	[2.0, 9.4]
7%	NPV ($B)	[-9.7, 78.3]	[2.8, 103.8]	[10.3, 119.7]	[15.1, 130.0]
	BCR	[-0.5, 4.1]	[0.1, 5.4]	[0.5, 6.2]	[0.8, 6.8]

Notes: Each cell reports the $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ bound across 50 fixed-cost draws. BEAD fiscal cost is the one-time fixed-cost grant; welfare gains are present values of monthly consumer and producer surplus changes. BCR = present-value welfare gain / fiscal cost. Negative lower-bound NPVs occur at high discount rates and short horizons when low fixed-cost draws produce minimal entry.

At a 3 percent discount rate over 20 years—a conservative assumption given that fiber infrastructure typically has a 25-to-30-year useful life—the BCR remains 2.4–9.7. Even at a 7 percent rate over 25 years, the lower bound on BCR is 0.5: only in the most pessimistic scenario (low fixed-cost draw, few marginal markets activated, high discount rate, short horizon) does the program fail the benefit-cost test. These findings are robust to the choice of discount rate in the plausible range. The main uncertainty comes from the width of the fixed-cost bounds rather than from the financial assumptions.

For ACP, the BCR calculation is less sensitive to the discount rate because both benefits and costs are recurring flows that do not require discounting relative to each other. The relevant sensitivity dimension for ACP is instead the assumed share of infra-marginal subscribers. If 50 percent of ACP recipients are infra-marginal (subscribe regardless of the discount), the effective BCR is approximately 0.5× the gross BCR in Table 17, ranging from 0.32–0.69 for the $30 discount. If only 20 percent are infra-marginal, the adjusted BCR rises to 0.50–1.10, spanning 1 in the upper bound. Thus ACP's fiscal efficiency depends on its ability to reach marginal subscribers.

Policy synthesis. The structural evidence points to a clear allocation principle. In markets with genuine deployment gaps—rural, low-density, and currently unserved tracts—the BEAD fixed-cost subsidy generates the highest welfare return per dollar, because the binding constraint is the absence of infrastructure, not its price. In markets where infrastructure already exists but affordability limits adoption—served urban and suburban tracts with many low-income households—ACP's demand-side discount delivers larger and more immediate welfare gains at substantially lower fiscal cost. No single instrument dominates across all markets: the programs are first-best responses to different market failures. The geographic complementarity documented in Section 6.3 has a direct implication for program design: a successor to ACP combined with continued BEAD deployment would address both failures simultaneously, reaching households that neither program alone can serve. Aggressive outreach and eligibility verification in any successor affordability program remain important for maximizing the share of transfers that reach marginal rather than infra-marginal subscribers.

8 Concluding Remarks

This paper evaluates ACP and BEAD using event study identification and a structural IO model of the U.S. broadband market with endogenous entry and product positioning. The event study documents reduced-form market responses to each program. The structural model then puts welfare numbers on the demand and entry channels it is built to capture and allows counterfactual policy comparisons.

ACP was associated with lower prices, upgraded product offerings, and lower provider concentration in states with higher eligibility exposure. BEAD's effects materialized with a two-to-three year lag and operated primarily through entry rather than product repositioning. In the model, a 75 percent BEAD subsidy generates 115,000–341,000 new provider-tract entries and a BCR of 3.5–12.7. A $30 ACP discount raises consumer surplus by $3.1–$3.3 billion per month at a BCR of 1.66–4.04. The fiscal gap reflects both the durability of infrastructure benefits and ACP's large infra-marginal transfers. Two caveats matter: the model is static and cannot be validated against post-program outcomes; and gross prices are held fixed in the ACP counterfactual.

ACP and BEAD solve different market failures and are economic complements. ACP addresses price-constrained adoption where infrastructure exists; BEAD addresses the deployment gap where private investment has not materialized. Neither alone closes the digital divide. BEAD's distributional gains are concentrated in the most underserved rural and minority-majority communities; ACP's are concentrated in served but low-income urban areas. Evaluations that count only new connections may miss broader price, product, and market-structure responses in oligopolistic markets. The structural counterfactuals here keep those responses separate from the modeled ACP discount. The choice between supply-side and demand-side intervention determines who benefits, through what channel, and at what fiscal cost.

With ACP exhausted and BEAD deployment still underway, the near-term policy question is whether a successor affordability program can be designed to reach marginal subscribers efficiently while limiting transfers to infra-marginal households. The structural framework developed here—which separately identifies the demand and entry channels and quantifies the welfare return to each—provides a toolkit for evaluating such designs ex ante rather than only after implementation.

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A Industry Background

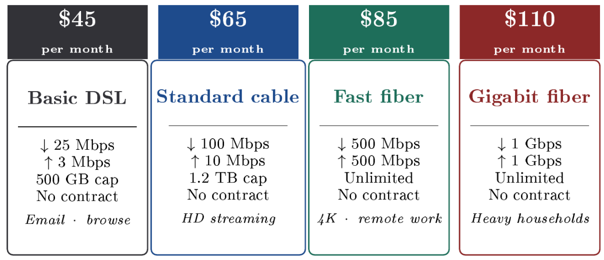

Figure A.1: Plan differentiated by speed, allowance, and price

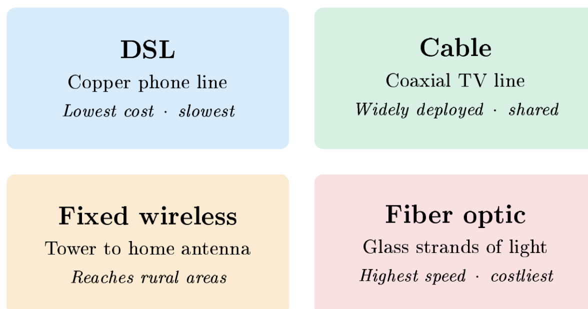

Figure A.2: Main broadband technologies

B Data

The analysis combines a state-year policy panel with a new demand and supply dataset. Table B.1 summarizes the data sources and their role in the paper.

Table B.1: Main Data Sources

Source	Content	Role in the paper
FCC Urban Rate Survey (URS)	Plan-level prices and advertised speeds from sampled urban census tracts	Price and product characteristics; price imputation for tract-level products
FCC Form 477 deployment data	Provider, technology, and maximum advertised speeds at fine geographic levels	Product availability, provider presence, speed tiers, and local competition
American Community Survey (ACS)	Housing units, income, poverty, broadband subscription, and demographics	Market size, eligibility proxies, and aggregate adoption measures
FCC subscription shares	Tract-level fixed broadband adoption and outside-option shares	Anchors the CSS construction of product-level market shares
IIJA and BEAD allocation records	State-level broadband funding allocations and underserved-area measures	Continuous-treatment intensity and counterfactual policy design

C Provider Normalization

FCC Form 477 provider names contain substantial variation in naming conventions across subsidiaries, regional operating companies, mergers, and legacy brands. I consolidate provider names into canonical provider identifiers representing the underlying corporate family. The normalization proceeds in two stages.

Generic Cleaning Rules

Provider names not matched through the override table are normalized using the following sequence:

1. Remove "d/b/a" suffixes;
2. Remove geographic qualifiers (e.g., "of Alabama");
3. Remove legal suffixes (LLC, Inc., Corp., Ltd., LP);
4. Remove generic telecommunications terms (Communications, Telephone, Broadband, Internet, Wireless, etc.);
5. Remove punctuation and collapse whitespace;

6. Retain the first two tokens as the canonical identifier.

D Additional Details on URS Price Assignment

Prices are assigned using plan-level observations from the FCC Urban Rate Survey. Let product (f, j, t) denote provider f, speed tier j, and tract-year market t, with average advertised download speed $\bar{d}_{fjt}$.

For each product, the algorithm searches for comparable URS plans using the following hierarchy:

1. State × provider × tier
2. National provider × tier
3. State × provider (any tier)
4. National provider (any tier)
5. State × tier (any provider)
6. National tier (any provider)

Within the first non-empty match category, the assigned price is the URS plan minimizing the absolute difference in advertised download speed:

$$m^* = \arg\min_{m \in \mathcal{M}_{fjt}} |d_m - \bar{d}_{fjt}|, \tag{D.1}$$

where d_m denotes URS plan download speed. All prices are converted to constant 2025 dollars using CPI-U adjustments.

Table D.1 reports the empirical distribution of match quality for the December 2020 estimation sample.

Table D.1: URS Price Imputation Match Quality

Match level	Products	Share
State $\times$ provider $\times$ tier	274,749	40.8%
National $\times$ provider $\times$ tier	103,288	15.3%
State $\times$ provider (any tier)	32,346	4.8%
National $\times$ provider (any tier)	20,001	3.0%
State $\times$ tier (any provider)	233,549	34.7%
National $\times$ tier (any provider)	9,177	1.4%
Total	673,110	100.0%

E CSS Market Share Construction

Product-level broadband shares are not directly observed. I construct them using a latent-share approach in the spirit of Crawford et al. [2019]. The method has two steps.

Step 1: Aggregate OLS logit. At the tract level, I estimate

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t'\kappa + D_t'\lambda + u_t, \tag{E.1}$$

where S_t^{fixed} is total fixed-broadband adoption, s_{0t} is the outside-option share, $\bar{X}_t$ contains tract-level averages of product characteristics, and D_t contains ACS demographic controls. The CSS estimation sample covers 70,990 tracts and has $R^2 = 0.253$.

Table 3 in Section 2 reports aggregate adoption moments for the December 2020 estimation sample. Table E.2 below reports the model-implied allocation of subscribers across providers. This allocation is useful as a diagnostic for the share-construction procedure, but it should not be interpreted as observed provider-level subscriber counts.

Step 2: Product-level allocation. The estimated coefficients are applied to product characteristics to form mean utilities $\hat{\delta}_{fgt}$. Product shares are allocated within each tract by softmax:

$$\hat{s}_{f|\text{in},t} = \frac{\exp(\hat{\delta}_{fgt})}{\sum_{f',g' \in \mathcal{F}_t} \exp(\hat{\delta}_{f'g't})}, \tag{E.2}$$

$$\hat{s}_{fgt} = \hat{s}_{f|\text{in},t} S_t^{\text{fixed}}. \tag{E.3}$$

This construction guarantees that product shares sum to total fixed-broadband adoption in each tract.

Table E.1: CSS Aggregate Simple Logit: Tract-Level OLS

Variable	Coefficient
Constant	-7.491^{***}
Average price / \$100	-0.279^{***}
Average download speed (Gbps)	0.314^{***}
Average download speed^2	-0.156^{***}
Average upload speed (Gbps)	0.034
Average upload speed^2	0.098^{***}
Average high-speed tier share	0.168^{***}
Average coverage share	0.471^{***}
Log household income	0.793^{***}
Share renting	0.401^{***}
Share college-educated	-0.052^{**}
Share aged 65+	0.274^{***}
N (tracts)	70,990
R^2	0.253

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$. Dependent variable is $\log(S_t^{\text{fixed}}/s_{0t})$.

Table E.2: Model-Implied Subscriber Allocation by Provider

Provider	Implied subscribers (Millions)	Share of inside good (%)	Share of households (%)	Tracts
AT&T	27.49	28.21	23.69	33,344
Comcast	17.28	17.74	14.90	28,950
Verizon	10.53	10.80	9.07	12,070
Frontier	7.83	8.03	6.75	9,549
Cox	3.77	3.87	3.25	5,540
Windstream	1.99	2.04	1.71	2,946
Consolidated	1.26	1.29	1.08	2,035
Starry	1.16	1.19	1.00	2,671
Optimum	1.13	1.16	0.98	2,670
All others (<1% each)	25.02	25.67	21.56	—

Notes: Provider shares are model-implied allocations based on tract-level adoption, observed provider availability, and product characteristics. They should not be interpreted as reported provider-level subscriber counts. “Share of inside good” is the provider’s share among inferred fixed-broadband subscribers; “Share of households” is relative to total Census households in the structural sample. All figures reflect the December 2020 baseline, which predates major IIJA broadband programs.

F Reduced-Form Robustness: Static DiD Estimates

Table F.1 reports static two-way fixed-effects DiD estimates for all thirteen outcomes. These specifications omit state-specific linear trends and are presented as robustness to the event study evidence in Section 3. For ACP, the static DiD without state trends yields large and significant coefficients on prices and market structure, reflecting pre-existing trend differences in broadband outcomes across states with different poverty rates rather than a causal ACP effect, as evidenced by the event study pre-period estimates in Figure 3. For BEAD, the static DiD collapses the lagged dynamics visible in Figure 4 into a single post-treatment average.

Table F.1: Difference-in-Differences Estimates: ACP and BEAD Exposure

Outcome	(1) ACP Elig. × Post_{ACP}	(2) BEAD log BEAD × $\text{Post}_{\text{BEAD}}$
Prices		
Log avg. real price	-0.5029^{**}	-0.0516^{*}
	(0.2228)	(0.0286)
Log price (<25/3 Mbps)	-0.2468^{**}	-0.0458^{*}
	(0.1119)	(0.0277)
Log price (≥25/3 Mbps)	-0.9902^{***}	-0.0717
	(0.2437)	(0.0506)
Technology & Speed		
Upload speed (Gbps)	-0.6422^{***}	-0.0373
	(0.1396)	(0.0536)
Fiber share	$+0.3467^{***}$	$+0.0225$
	(0.0611)	(0.0195)
DSL share	-0.4419^{***}	-0.0171
	(0.0546)	(0.0295)
Fixed-wireless share	-0.0663	$+0.0175$
	(0.0573)	(0.0229)
Market Structure		
Provider count	$+1.9557$	-0.2484
	(3.2402)	(0.3961)
Provider × technology count	$+1.5790$	-0.5047
	(4.3823)	(0.5050)
Provider entries	$+1.6016$	$+0.3147$
	(3.4076)	(0.2054)
Provider exits (next year)	$+7.9843^{***}$	$+0.0557$
	(1.7082)	(0.1695)
Provider HHI	-0.1483^{**}	-0.0355^{*}
	(0.0591)	(0.0213)
Top-provider share	-0.2114^{***}	-0.0458^{*}
	(0.0713)	(0.0258)
Observations	597	597
State FE	Yes	
Year FE	Yes	
Weights	Number of plans	

Notes: Each cell reports the OLS coefficient and standard error (in parentheses) from a separate two-way fixed-effects regression. The ACP treatment is the share of households below 200% of the federal poverty level interacted with an indicator for years ≥ 2022. The BEAD treatment is the log of BEAD allocation per unserved location interacted with an indicator for years ≥ 2023. Standard errors are clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

G Measurement and Limitations

The structural analysis rests on several measurement assumptions that are worth stating clearly, as they affect how the estimates should be interpreted.

Inferred product-level market shares. The most significant data limitation is the absence of directly observed product-level subscriber counts. The FCC reports where broadband services are available and aggregate adoption rates by speed threshold, but not the distribution of subscribers across competing ISPs within a census tract. I construct product-level shares using the CSS procedure of Crawford et al. [2019], which allocates tier-level adoption across providers according to logit shares implied by product characteristics. This procedure is internally consistent and produces share distributions consistent with aggregate adoption targets, but it imposes a particular allocation mechanism that may not match actual subscriber distributions. If some providers have disproportionately high or low local market shares not captured by the characteristic-based allocation, the demand estimates will reflect the average allocation rule rather than true subscriber counts. This is a recognized limitation of demand estimation in markets without scanner or administrative data, and it parallels similar assumptions in related structural work [Crawford et al., 2019, Fan, 2013].

Imputed URS prices. Prices are assigned using the FCC Urban Rate Survey rather than actual transaction prices. The URS reports plan-level advertised prices from a stratified sample of urban census tracts, which may not perfectly reflect prices in rural or suburban markets. For approximately 35 percent of products, prices are assigned at the state-tier level rather than at the more granular brand-tier-state level. To the extent that URS prices differ systematically from prices actually charged in these tracts, the demand estimates will be attenuated or biased in a direction that depends on the nature of the mismatch. The BLP IV strategy mitigates this by using rivals' characteristics rather than own prices as instruments, but attenuation from price measurement error in the dependent variable remains a potential concern.

Alternative aggregation as robustness. To check robustness of the demand estimates to the share-construction procedure, an earlier version of the analysis estimated demand using a state-by-year aggregation approach. In that alternative, market shares were constructed by combining ACS aggregate broadband demand with plan-availability measures at the state level, under the assumption that plan availability was proportional to aggregate demand allocation within states. This approach produced mean own-price elasticities around -5.87, reasonably close to the current tract-level estimate of -6.33. The similarity provides some reassurance that the elasticity magnitudes are not entirely driven by the tract-level share-construction procedure, though the alternative approach has its own identification limitations.

Static model with December 2020 baseline. The model is static: it characterizes the equilibrium in December 2020 and uses this to simulate counterfactual policies. It does not model dynamic investment decisions, learning, or network effects. The December 2020 baseline predates ACP and BEAD implementation, which is important for the counterfactuals but means the model cannot be directly validated against post-program outcomes. These are recognized limitations common to static structural models of entry in infrastructure industries, and they are acknowledged when interpreting the counterfactual results.

H Demand Specification Details

The demand covariate vector includes download speed, download speed squared, upload speed, upload speed squared, and a high-speed tier indicator. The excluded instruments are mean rivals' download speed, mean rivals' upload speed, mean rivals' high-speed share, and the number of same-tier rivals. State fixed effects absorb regulatory and demographic differences across states. Provider $\times$ Tier fixed effects absorb permanent provider-tier quality differences. Fixed effects are absorbed by two-way iterative demeaning with tolerance 10^{-10}.

Table 8 in Section 5 reports the full first-stage diagnostics for the preferred 2SLS-FE specification. The Angrist-Pischke F-statistics of 1,173.9 (price) and 195,007.6 (within-share) far exceed the Stock and Yogo [2002] threshold of 10. The Wooldridge score test for overidentifying restrictions yields a statistic of 14.84 with $p = 0.001$. With more than 216,000 observations, the test has high power against small deviations from exact exclusion; I interpret the instruments as strong but plausibly approximate.

I Markup Distributions and Diagnostics

Tables 10 and 11 in Section 5 report the full distribution of Bertrand-Nash markups, marginal costs, and Lerner indices by speed tier and firm type. Figure 6 in the same section plots the empirical distributions of structural residuals: demand shocks, marginal-cost shocks, Bertrand-Nash markups, and price-to-marginal-cost ratios. Key findings are summarized here for reference.

Technology-type breakdown. Decomposing marginal costs by technology reveals that fiber products have significantly higher marginal costs than DSL (median \$68 vs. \$41 per month), consistent with the higher infrastructure costs of fiber relative to legacy copper networks. Cable products are intermediate (median \$55). Fixed-wireless products have the lowest marginal costs in the sample (median \$36), reflecting their lower capital intensity relative to fiber and cable. These technology differences partly explain the high-speed tier premium: fiber dominates Tier H, raising the average marginal cost of high-speed products relative to the low-speed DSL-dominated Tier L.

J Entry Moment Inequalities

This appendix gives the technical details behind the fixed-cost estimation and counterfactual product-choice algorithm. For each potential product position $(f, g) \in \mathcal{J}_t^0$, let

$$\Delta_{fgt}(Y_{-fgt}) = r_{ft}(Y_{fgt} = 1, Y_{-fgt}) - r_{ft}(Y_{fgt} = 0, Y_{-fgt}) \tag{J.1}$$

be the change in provider f's variable profit from adding tier g in tract t.

Panel construction. The estimation panel is the balanced set of tract-provider-tier positions defined by the county-footprint rule in (4.11). The outcome is $Y_{fgt} = 1$ for observed provider-tier products and zero for feasible but unobserved positions. The latent fixed cost is

$$FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0, 1), \tag{J.2}$$

where $\theta_{b(t),g}$ is the systematic fixed-cost component for market-size cell $b(t)$ and tier g.

Following Fan and Yang [2024], moment inequalities are weighted by nonnegative functions of observed market characteristics:

$$h^{(k)}(X_t, g) \in \left\{ \begin{array}{l} \mathbf{1}\{b(t) = \text{Small}, g = H\}, \mathbf{1}\{b(t) = \text{Small}, g = L\}, \mathbf{1}\{b(t) = \text{Medium}, g = H\} \\ \mathbf{1}\{b(t) = \text{Medium}, g = L\}, \mathbf{1}\{b(t) = \text{Large}, g = H\}, \mathbf{1}\{b(t) = \text{Large}, g = L\} \end{array} \right\}.$$

Assumption 4. *For each potential position (f, g, t), the observed decision Y_{fgt} is not a dominated strategy.*

This implies

$$\Pr(Y_{fgt} = 1 \text{ is dominant}) \leq \Pr(Y_{fgt} = 1) \leq \Pr(Y_{fgt} = 1 \text{ is not dominated}). \tag{J.3}$$

The lower and upper incremental-profit cutoffs are

$$\underline{\Delta}_{fgt} = \min_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}), \tag{J.4}$$

$$\overline{\Delta}_{fgt} = \max_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}). \tag{J.5}$$

The implied conditional inequalities are

$$\Phi\left(\frac{\underline{\Delta}_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right) \leq \Pr(Y_{fgt} = 1 \mid X_t, g), \tag{J.6}$$

$$\Pr(Y_{fgt} = 1 \mid X_t, g) \leq \Phi\left(\frac{\overline{\Delta}_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right). \tag{J.7}$$

Criterion and confidence set. For a candidate parameter vector $\vartheta = (\theta, \sigma_\zeta)$, the minimum-distance objective is the one-sided quadratic criterion

$$Q_n(\vartheta) = \left[\sqrt{n}\,\max\{\hat{m}(\vartheta), 0\}\right]' \hat{W}_n \left[\sqrt{n}\,\max\{\hat{m}(\vartheta), 0\}\right], \tag{J.8}$$

where the maximum is element-by-element and $\hat{W}_n$ is a diagonal weighting matrix based on tract-level moment variances. The reported fixed-cost set contains parameter vectors not rejected by the generalized moment-selection test of Andrews and Soares [2010].

Drawing fixed costs. For each accepted ϑ, the latent fixed cost is $FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}$. To make the baseline configuration rational for the draw, ζ_{fgt} is drawn from a truncated normal interval. For observed positions,

$$\zeta_{fgt} \leq \frac{\Delta_{fgt}(Y^{obs}_{-fgt}) - \theta_{b(t),g}}{\sigma_\zeta},$$

and for unobserved feasible positions,

$$\zeta_{fgt} > \frac{\Delta_{fgt}(Y^{obs}_{-fgt}) - \theta_{b(t),g}}{\sigma_\zeta}.$$

Counterfactual best responses. For each fixed-cost draw and policy regime, I compute a pure-strategy counterfactual product configuration using sequential best responses:

1. Initialize at $Y_t^{(0)} \in \{Y_t^{obs}, \mathbf{0}, \mathbf{1}\}$.

2. Given the current configuration, update firms sequentially. When firm f is updated, hold rivals' current choices fixed and solve

$$Y_{ft}^{BR} \in \arg \max_{Y_{ft} \in \{\varnothing, \{H\}, \{L\}, \{H,L\}\}} \left\{ r_{ft}(Y_{ft}, Y_{-ft}) - \sum_{g \in \{H,L\}} Y_{fgt} FC^{cf}_{fgt} \right\}.$$

3. Replace firm f's portfolio with Y_{ft}^{BR}.

4. Repeat until no firm changes its portfolio. Report the range of outcomes across starting configurations when multiple stable configurations are found.

K Fan and Yang Bounds

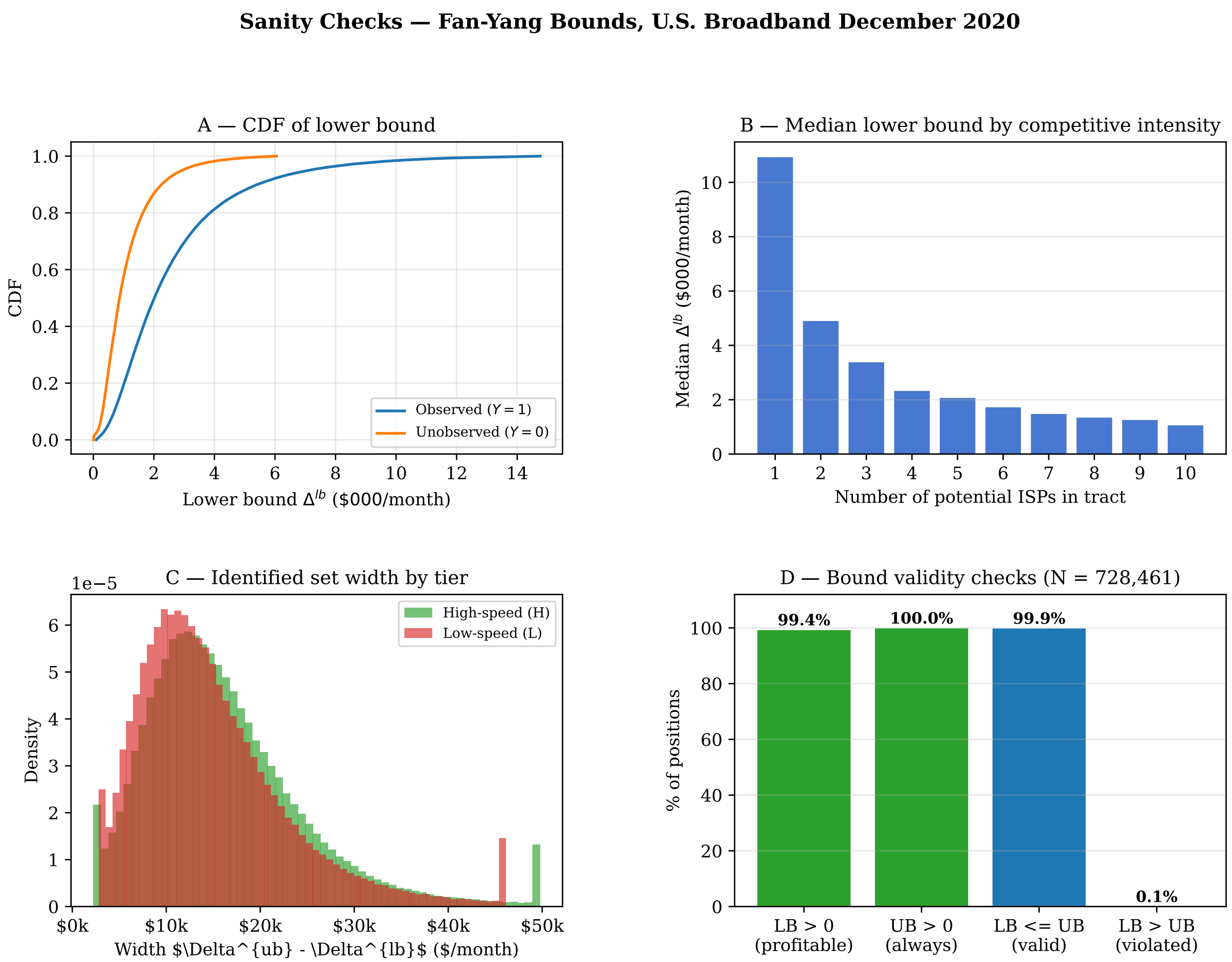

Figure K.1: Fan and Yang Bounds – Sanity Checks

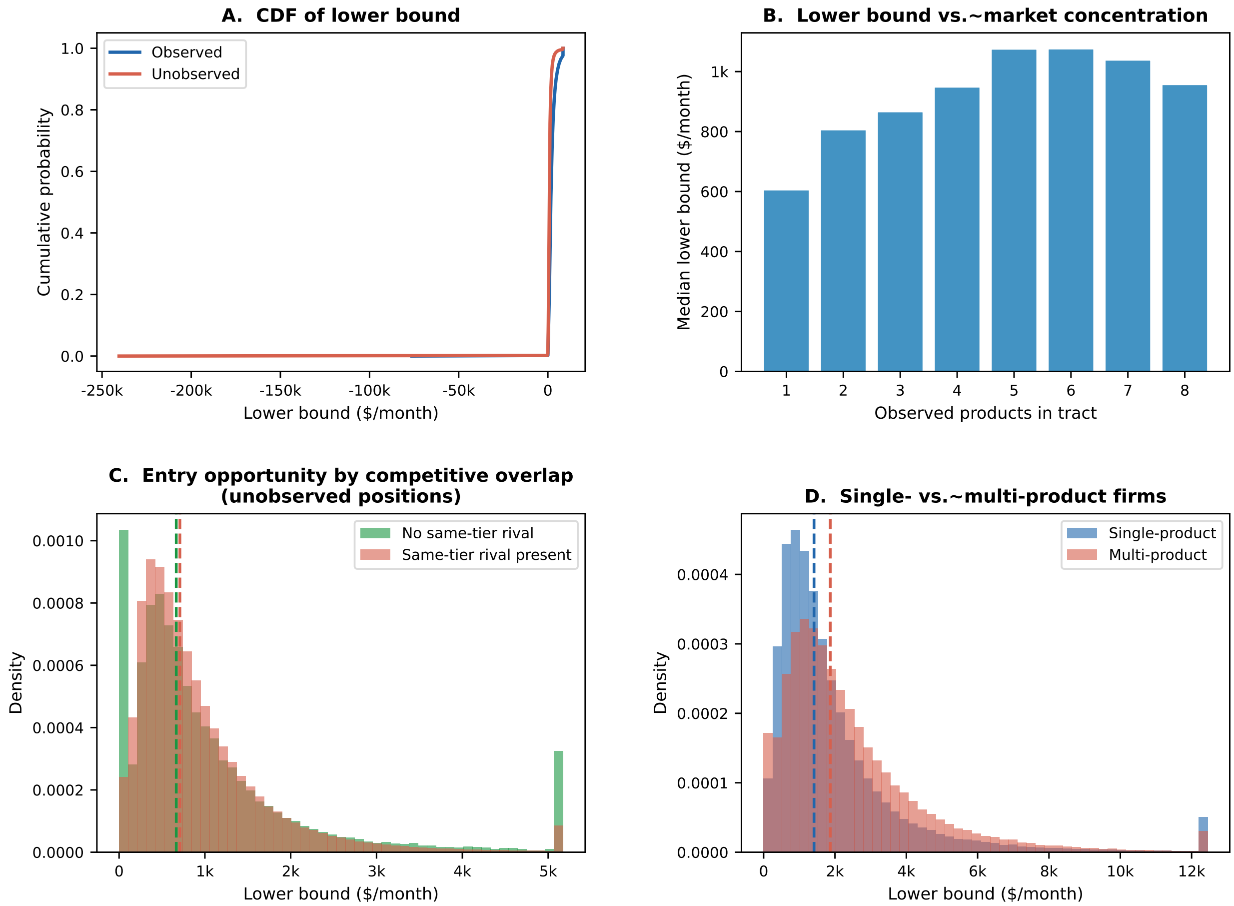

Figure K.2: Fan and Yang Bounds by Market Structure

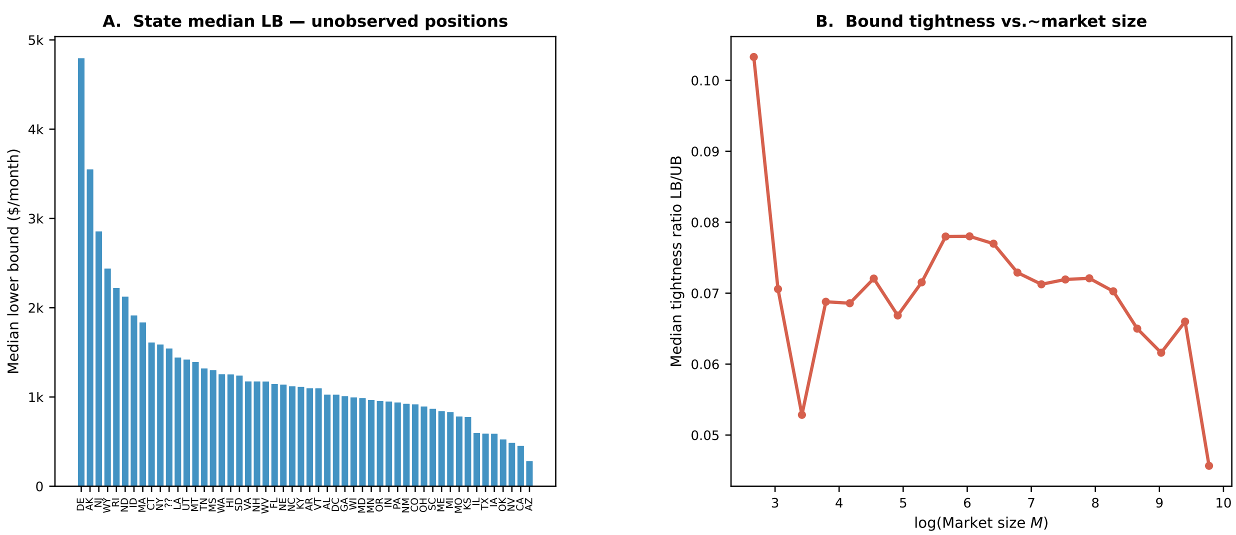

Figure K.3: Fan and Yang Bounds by Geographic areas and Market size

L Detailed Counterfactual Outcomes

Consumer surplus in tract t is computed from the inclusive value:

$$\mathrm{CS}_t(Y_t, p_t) = \frac{M_t}{-\hat{\alpha}/100} \log\left[1 + \left(\sum_{j\in\mathcal{J}_t(Y_t)} \exp\left(\frac{\hat{\delta}_{jt}}{1-\hat{\rho}}\right)\right)^{1-\hat{\rho}}\right], \tag{L.1}$$

where prices enter $\hat{\delta}_{jt}$ in dollars and the demand estimate uses price divided by 100. Producer surplus is variable profit net of fixed costs, and social surplus is the sum of consumer and producer surplus minus public program costs.

Discrete product-choice games may have multiple equilibria [Ciliberto and Tamer, 2009]. I initialize the algorithm at the observed baseline configuration, the empty configuration, and the full configuration. If starting values converge to the same product configuration, I use that as the counterfactual equilibrium. If they converge to different stable configurations, I report the range of counterfactual outcomes across equilibria. National lower and upper bounds are obtained by summing tract-level lower and upper outcomes.

M Cost-Benefit Robustness

Table M.1 reports the sensitivity of NPV bounds and benefit-cost ratios to alternative annual discount rates, holding the benchmark 25-year horizon fixed. The ACP benefit-cost ratio is unchanged by the discount rate because both benefits and costs are recurring annual flows.

Table M.1: Cost-Benefit Robustness at 25-Year Horizon

Policy scenario	1%	3%	5%	7%
Net present value bounds ($B)				
BEAD 25% FC reduction	[-40.3, 139.5]	[-32.7, 109.8]	[-27.2, 88.5]	[-23.1, 72.8]
BEAD 50% FC reduction	[1.8, 187.5]	[-0.7, 146.6]	[-2.4, 117.1]	[-3.7, 95.5]
BEAD 75% FC reduction	[80.9, 286.8]	[59.0, 222.8]	[43.3, 176.7]	[31.7, 142.8]
ACP $20 monthly discount	[35, 725]	[28, 573]	[23, 464]	[19, 383]
Benefit-cost ratio bounds				
BEAD 25% FC reduction	[-9.6, 64.4]	[-7.6, 50.9]	[-6.2, 41.2]	[-5.1, 34.1]
BEAD 50% FC reduction	[1.2, 24.7]	[0.9, 19.6]	[0.8, 15.8]	[0.6, 13.1]
BEAD 75% FC reduction	[4.4, 16.1]	[3.5, 12.7]	[2.8, 10.3]	[2.3, 8.5]
ACP $20 monthly discount	[1.17, 4.70]	[1.17, 4.70]	[1.17, 4.70]	[1.17, 4.70]

Notes: BEAD NPV rows report $\mathrm{PV}(\Delta SW) - G$. BEAD BCR rows report $\mathrm{PV}(\Delta SW)/G$. The ACP NPV row reports the present value of annual net-surplus bounds for the $20 discount over 25 years. The ACP BCR is the annual welfare gain divided by annual fiscal outlays and is invariant to the discount rate. The benchmark specification uses the 3 percent column.

Online Appendix

OA.1 Replication Pipeline

The demand, supply, and fixed-cost objects are generated by five scripts. Table OA.1.1 summarizes the pipeline.

Table OA.1.1: Demand and Supply Replication Pipeline

#	Script	Input	Output
1	`css_simple_logit_summary_2020.py`	FCC 477, URS, ACS	CSS implied shares
2	`demand_model_iv.py`	Implied shares	Tract-provider-tier collapsed data
3	`nested_logit_demand_2020.py`	Collapsed data	Demand estimates and Bertrand-Nash markups
4	`demand_diagnostics_2020.py`	Collapsed data and markups	Diagnostic plots and marginal-cost summary
5	`estimate_fixed_costs_fan_yang.py`	Collapsed data and profit cutoffs	Product-position panel and fixed-cost moments

OA.2 Raw Data Construction

FCC Form 477 deployment data are filtered to residential fixed broadband technologies: DSL, cable, fiber, and fixed wireless. Satellite and mobile technologies are excluded. URS prices are matched to products using a hierarchical match by provider, state, technology, and speed tier. ACS and FCC subscription files provide market size, outside-option shares, aggregate adoption, and demographic controls. The resulting demand file is collapsed to the tract × canonical-provider × speed-tier level.

OA.3 Variable Definitions

The high-speed tier is defined as download speed at least 25 Mbps and upload speed at least 3 Mbps. Prices are monthly subscription prices. Market size is the number of housing units. The outside option is the share of households without fixed broadband. The preferred demand specification uses State and Provider × Tier fixed effects; the preferred marginal-cost specification uses no fixed effects because speed-cost variation is primarily between providers and technologies.

OA.4 Implementation Notes

Two-way fixed effects are absorbed through iterative demeaning with tolerance 10^{-10} and a maximum of 500 iterations. Demand is estimated by 2SLS using heteroskedasticity-robust standard errors. Markups are recovered tract by tract by solving the Bertrand-Nash linear system in (4.8). Marginal costs are computed as prices minus recovered markups. Observations with non-positive implied marginal costs are excluded from the log marginal-cost regression but reported in the markup diagnostics.